GAZİ UNIVERSITY

FACULTY OF ENGINEERING
DEPARTMENT OF INDUSTRIAL ENGINEERING

Lecturer : Dr Ercan Ezin

IE326-DATABASE MANAGEMENT SYSTEMS

WEEK 4: LOGICAL DESIGN

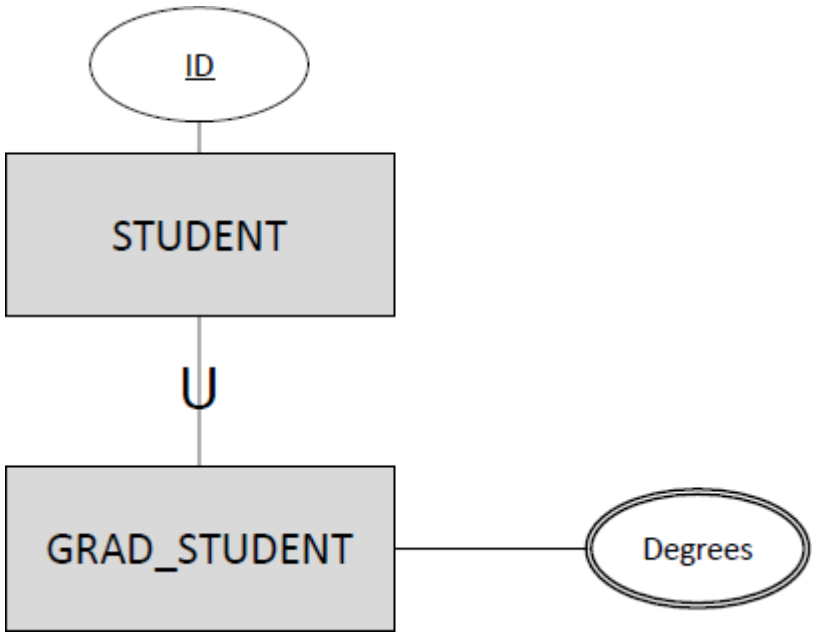
DATABASE DESIGN

CONCEPTUAL DESIGN (23 Feb-Presented)

LOGICAL DESIGN(Today's topic)

Specialization/Generalization

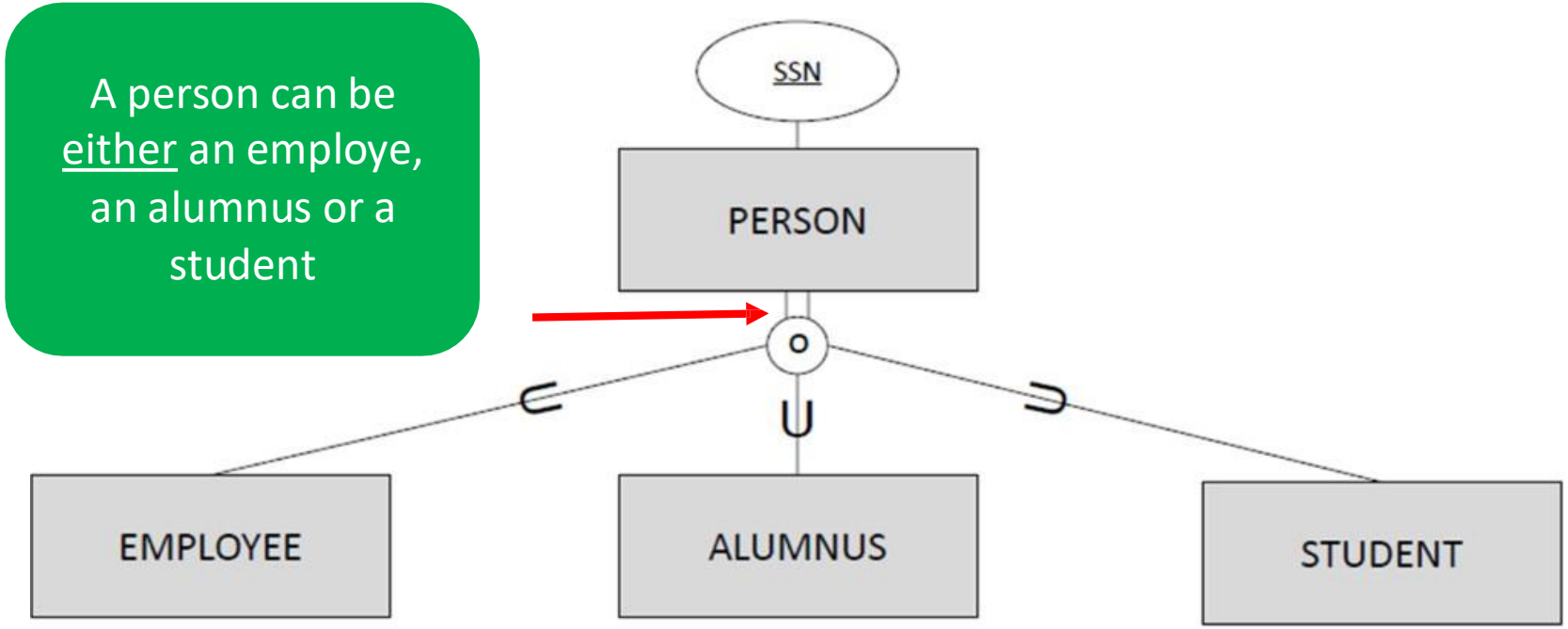
- Only a subset of entities within a type have certain attributes or participate in certain relationships



Multiple Subtypes: Overlapness

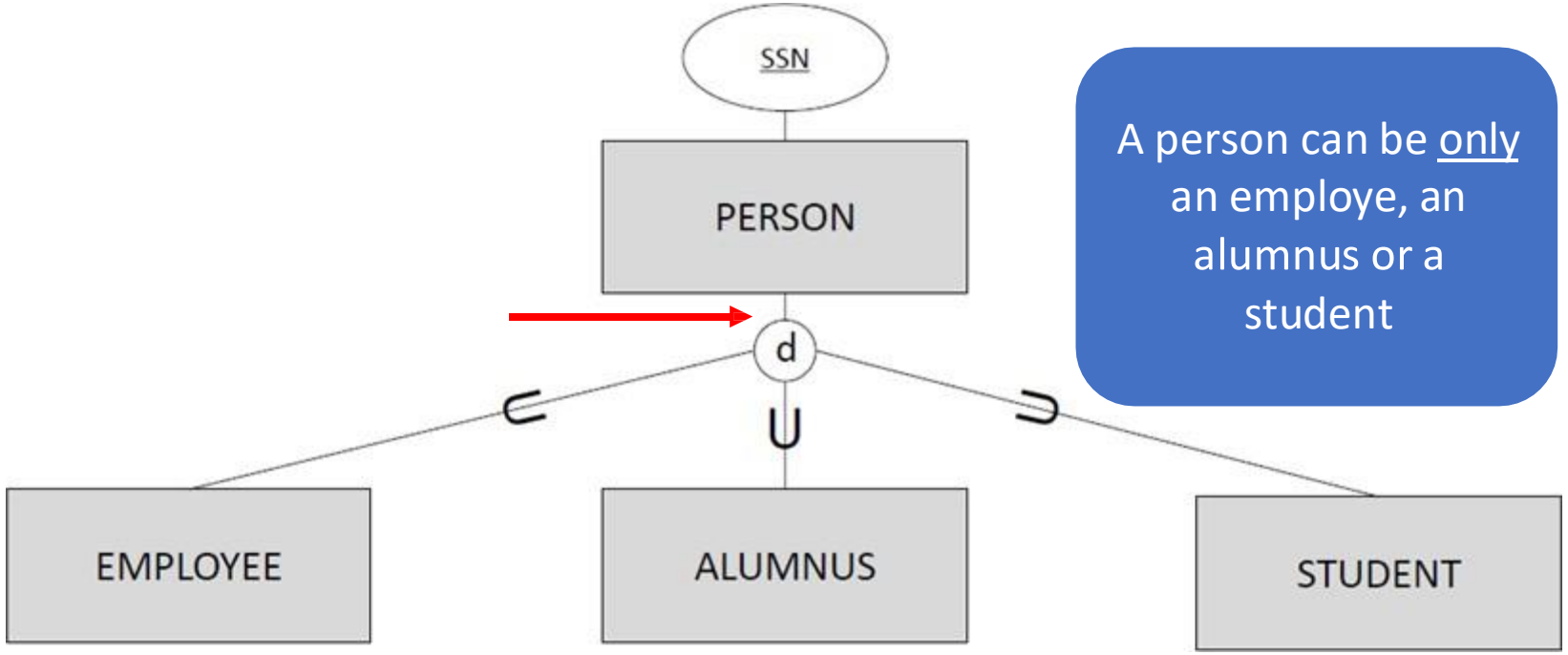
- (o)verlap: may be more than one
- (d)isjoint: entities may only be one subtype

A person can be <u>either</u> an employe, an alumnus or a student



Multiple Subtypes: Disjointedness

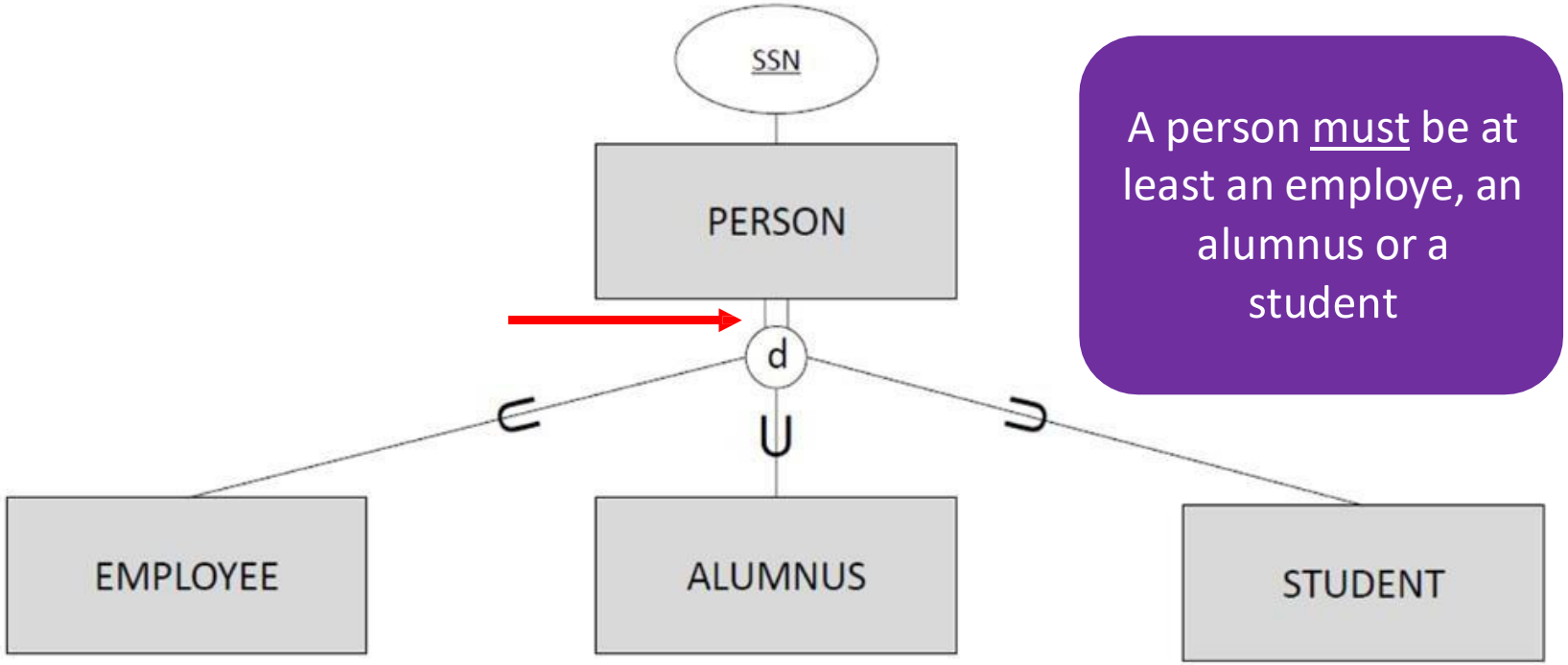
- (o)verlap: may be more than one
- (d)isjoint: entities may only be one subtype



A person can be <u>only</u> an employe, an alumnus or a student

Multiple Subtypes: Completeness

- Similar to relationships; can be total (must belong to subtypes) or partial (can belong)

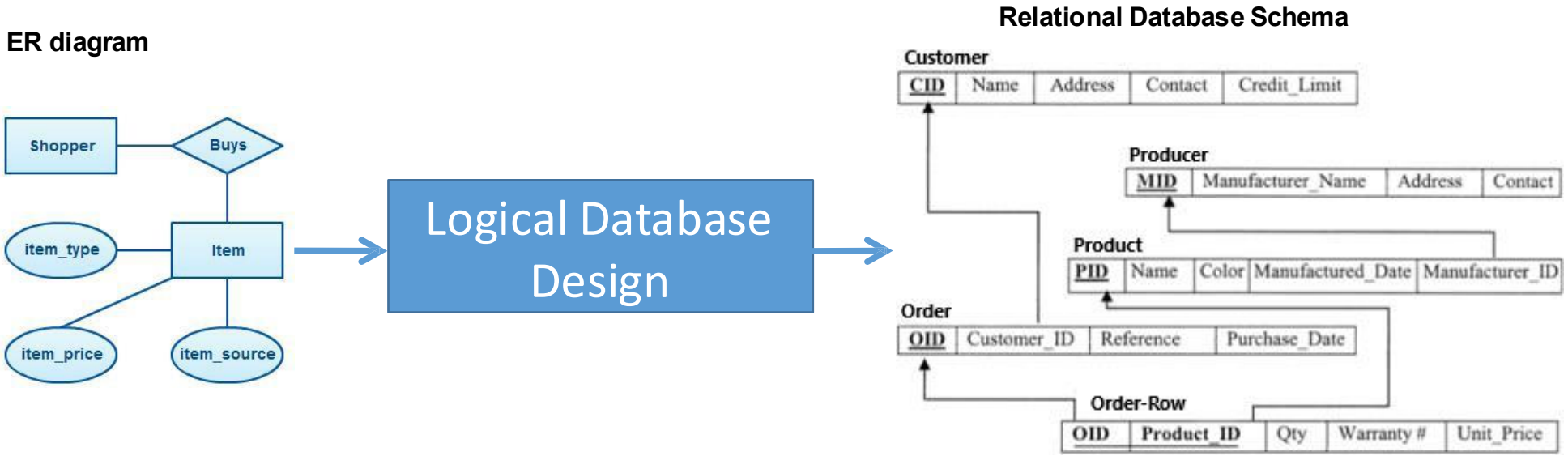


A person must be at least an employe, an alumnus or a student

LOGICAL DESIGN

Introduction

- Within the course scope, it will be assumed that DBMS has arelational model schema.



Relation and Relationship Chart

- The Schema that will be used in a Logical DB Design.

Example representation of a Logical DB:

Customer

Customer ID	First Name	Last Name	Telephone	Address

Relation and Relationship Chart

Some important features for relationships consisting of rows and columns are as follows:

- The relationship has a singular name.
- Attributes are in a simple attribute structure.
- The relation has at least one key.
- Attributes have singular nouns.
- Attribute order is unimportant.
- Row order is unimportant as well.

Converting ER Diagrams to Relationships

Basic Step → Converting the entities

- Consider only simple attributes.
- Place the schemas close to the ER diagrams.
- Set the necessary gaps for the connections.

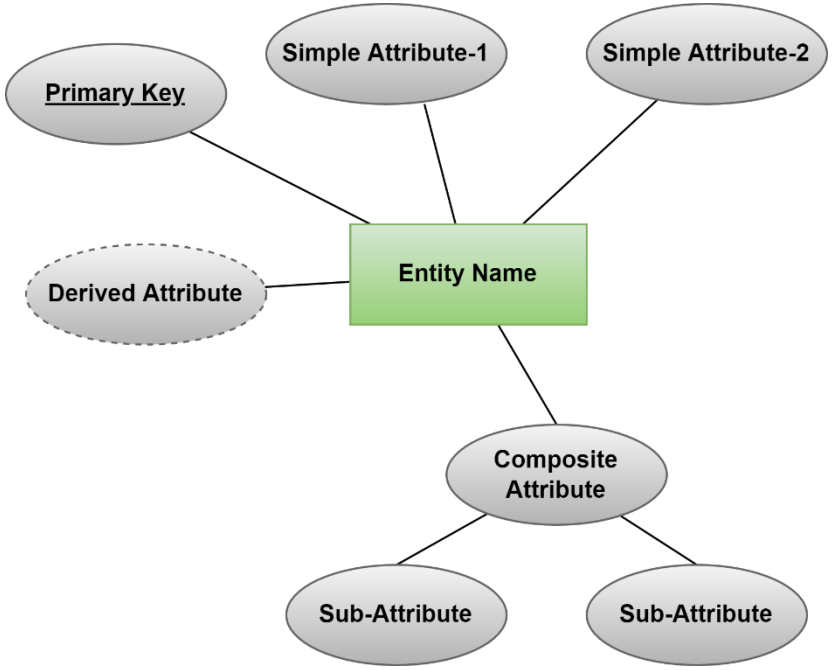
Basic Step → Converting the Relationships

- Pay attention to the relationship types.

Additional operation → Converting the Complex Attributes

- Multi-valued attributes
- Composite attributes

1. Conversion of Entities



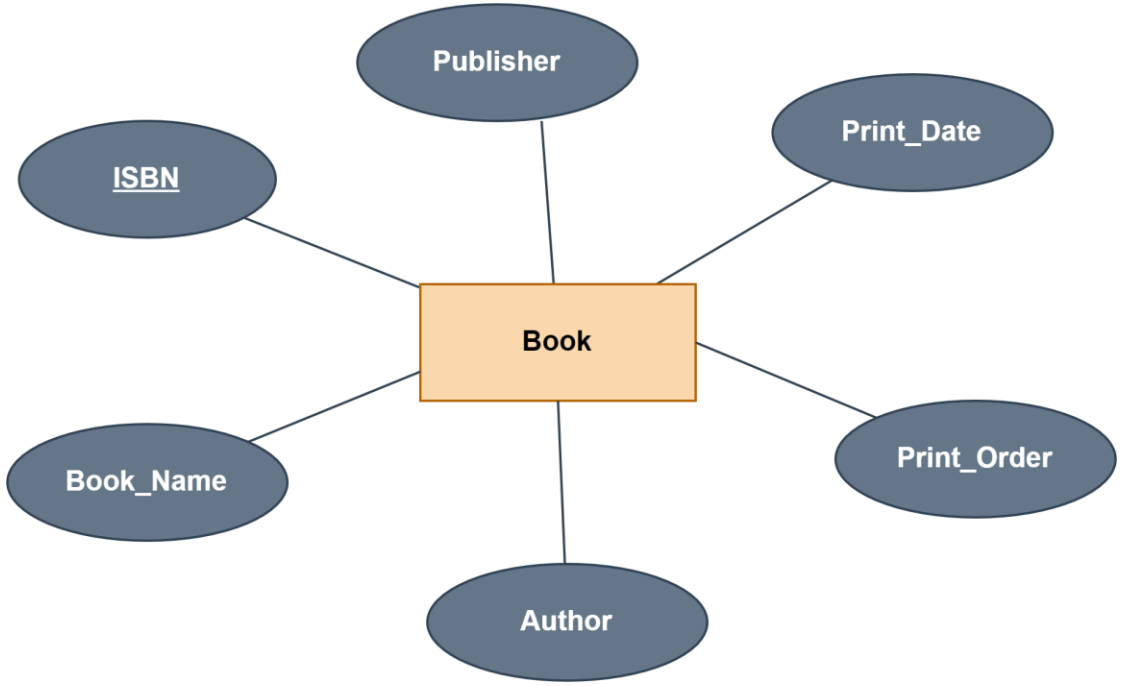
Rules to create a schema:

- Create an empty schema with the same name as the entity.
- Move the primary key of the entity to the schema.
- Move the simple attributes of the entity to the schema.
- Ignore the other attributes.

Entity Name

Primary Key	Simple Attribute-1	Simple Attribute-2

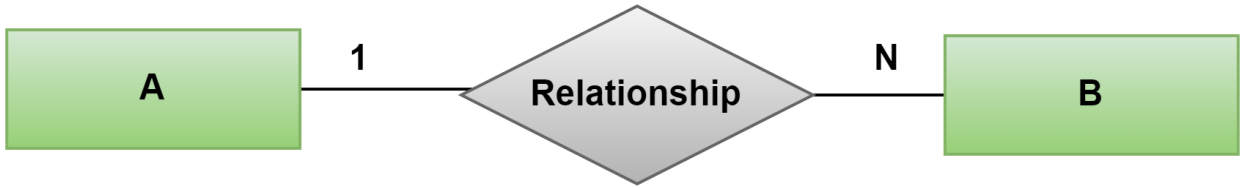
1. Conversion of Entities



Book

ISBN	Book_Name	Publisher	Author	Print_Order	Print_Date

2. Conversion of 1:N Binary Relationships



- Create a schema of Entity A with a singular relationship using known methods.
- Create a schema of Entity B that is on the side of multiple relationships (N).
- Add the primary key of Entity A to schema of Entity B as foreign key.
- Draw an arrow to the primary key starting from the foreign key.

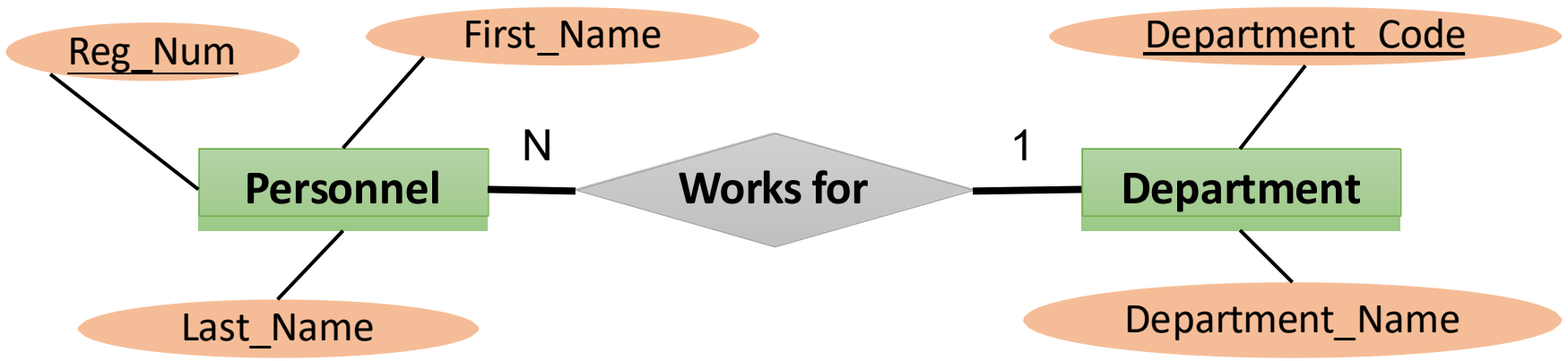
Entity A

Primary Key_A	....

Entity B

Primary Key_B	....	Foreign_Key_A

2. Conversion of 1:N Binary Relationships



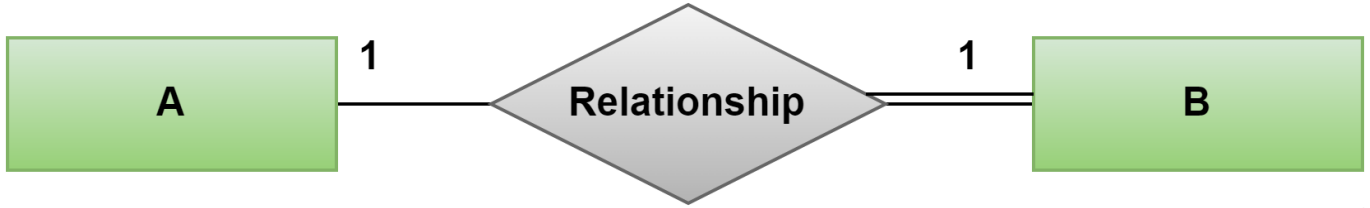
Department

Department_Code	Department_Name

Personnel

Reg_Num	First_Name	Last_Name	Department_Code

3. Conversion of 1:1 Binary Relationships



- Create two different relationship schemes with known methods for entities on both component sides.
- Add the primary key attribute of the entity with the mandatory join to the schema of the entity with the partial join relation as a foreign key.
- Draw arrow starting from the foreign key and ending at the primary key.

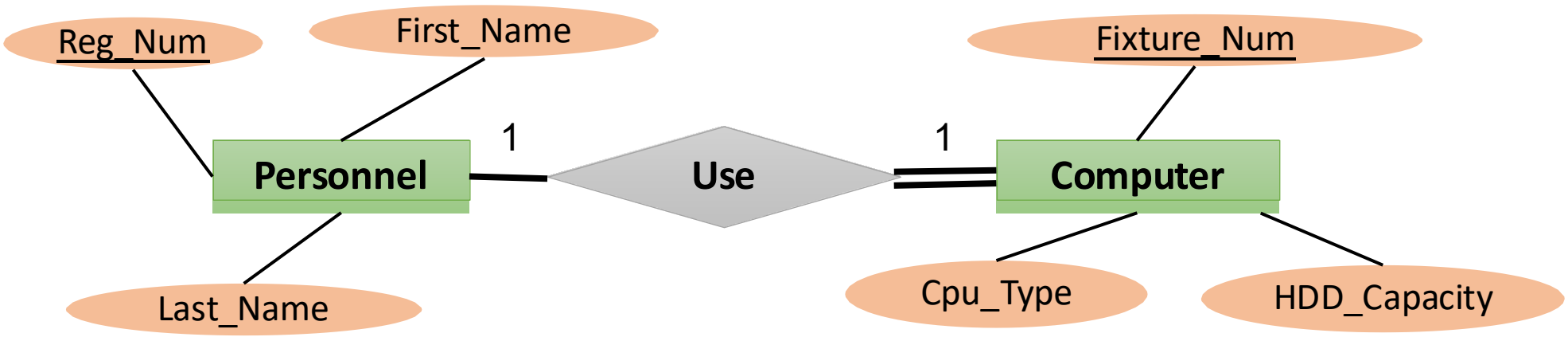
Entity A

Primary Key_A	....

Entity B

Primary Key_B	Foreign_Key_A

3. Conversion of 1:1 Binary Relationships



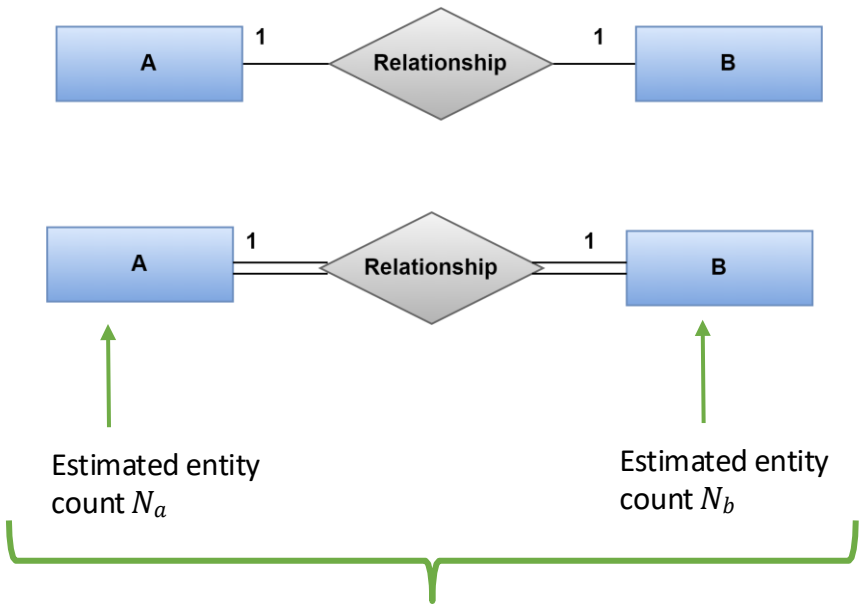
Personnel

Reg_Num	First_Name	Last_Name

Computer

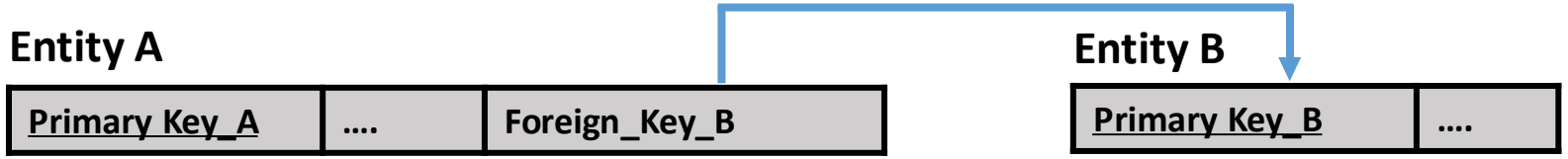
Fixture_Num	Reg_Num	Cpu_Type	HDD_Capacity

3. Conversion of 1:1 Binary Relationships

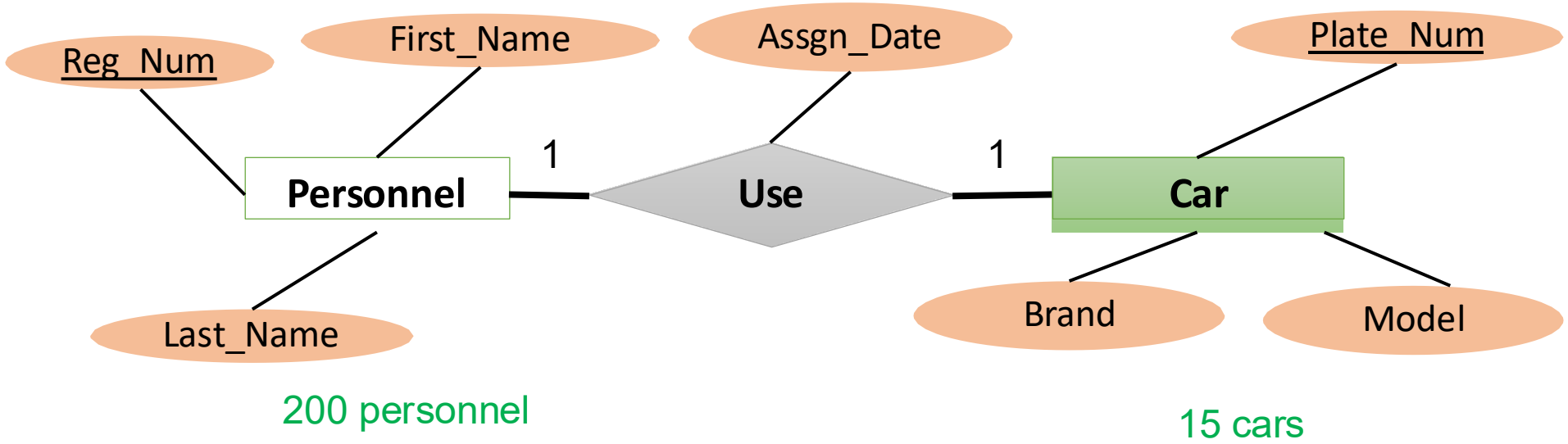


$N_a <= N_b$

- Create two different relationship schemes with known methods for entities on both component sides.
- Add the primary key of the other entity as a foreign key to the relationship schema of the entity with the smallest number of elements in the entity set.
- Draw arrow starting from the foreign key and ending at the primary key.



3. Conversion of 1:1 Binary Relationships



Personnel

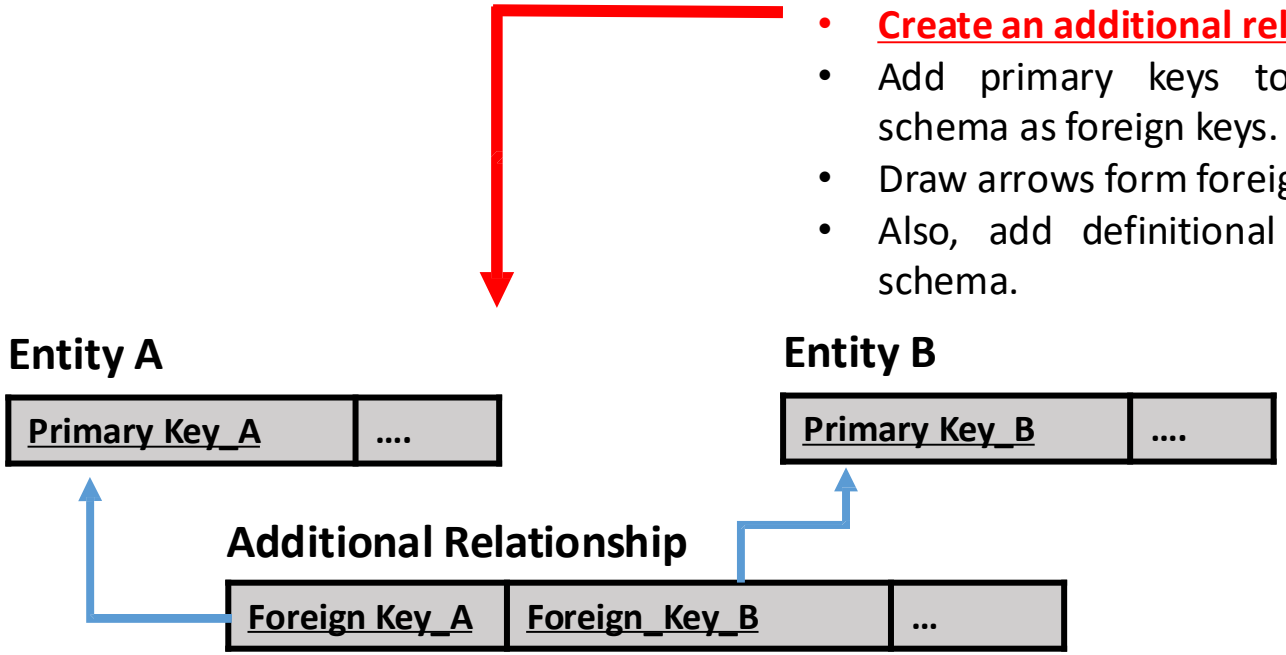
Reg_Num	First_Name	Last_Name

Car

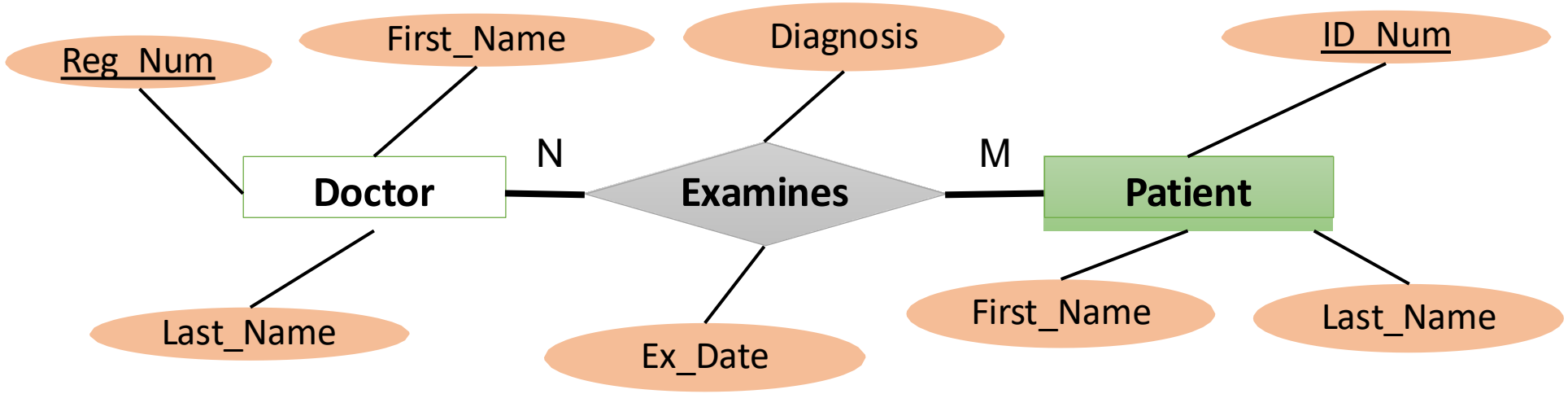
Plate_Num	Brand	Model	Reg_Num	Assgn_Date

4. Conversion of N:M Binary Relationships

- Create two different relationship schemes with known methods for entities on both component sides.
- **Create an additional relationship.**
- Add primary keys to the additional relationship schema as foreign keys.
- Draw arrows form foreign keys to primary keys.
- Also, add definitional attributes to the additional schema.



4. Conversion of N:M Binary Relationships



Doctor

Reg_Num	First_Name	Last_Name

Treatment

Reg_Num	ID_Num	Ex_Date	Diagnosis

Patient

ID_Num	First_Name	Last_Name

5. Conversion of Weak Entities

- The strong entity's relationship is established using known methods.
- An empty relationship schema is created for the weak entity.
- The primary key of the strong entities are added to the relationship scheme as the foreign key attribute.
- The key part attributes of the weak entity are also added to the schema, creating the schema's primary key.
- All other simple attributes remaining in the weak entity are added to the schema.

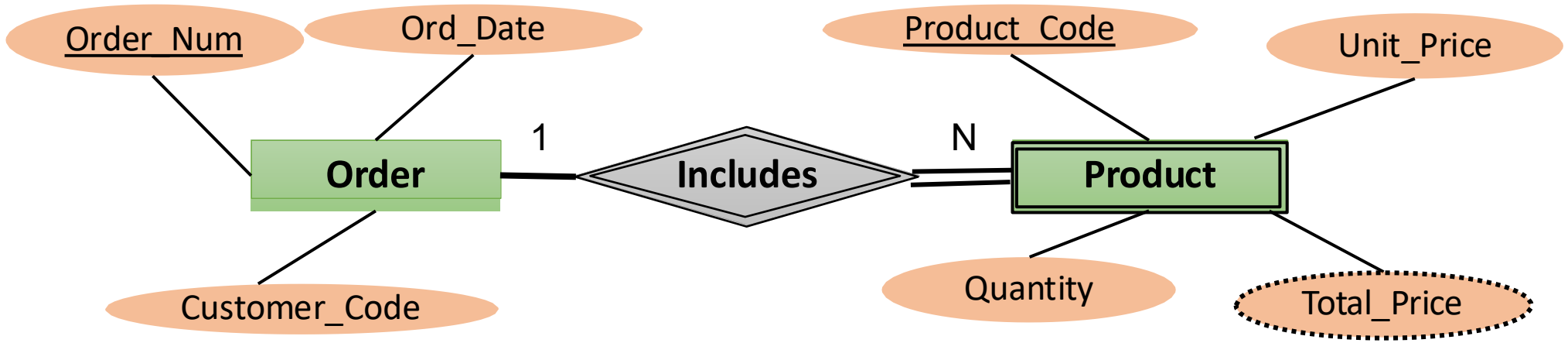
Entity A

Primary Key_A	....

Entity B

Foreign Key_A	Part_of_Key_B	...

5. Conversion of Weak Entities



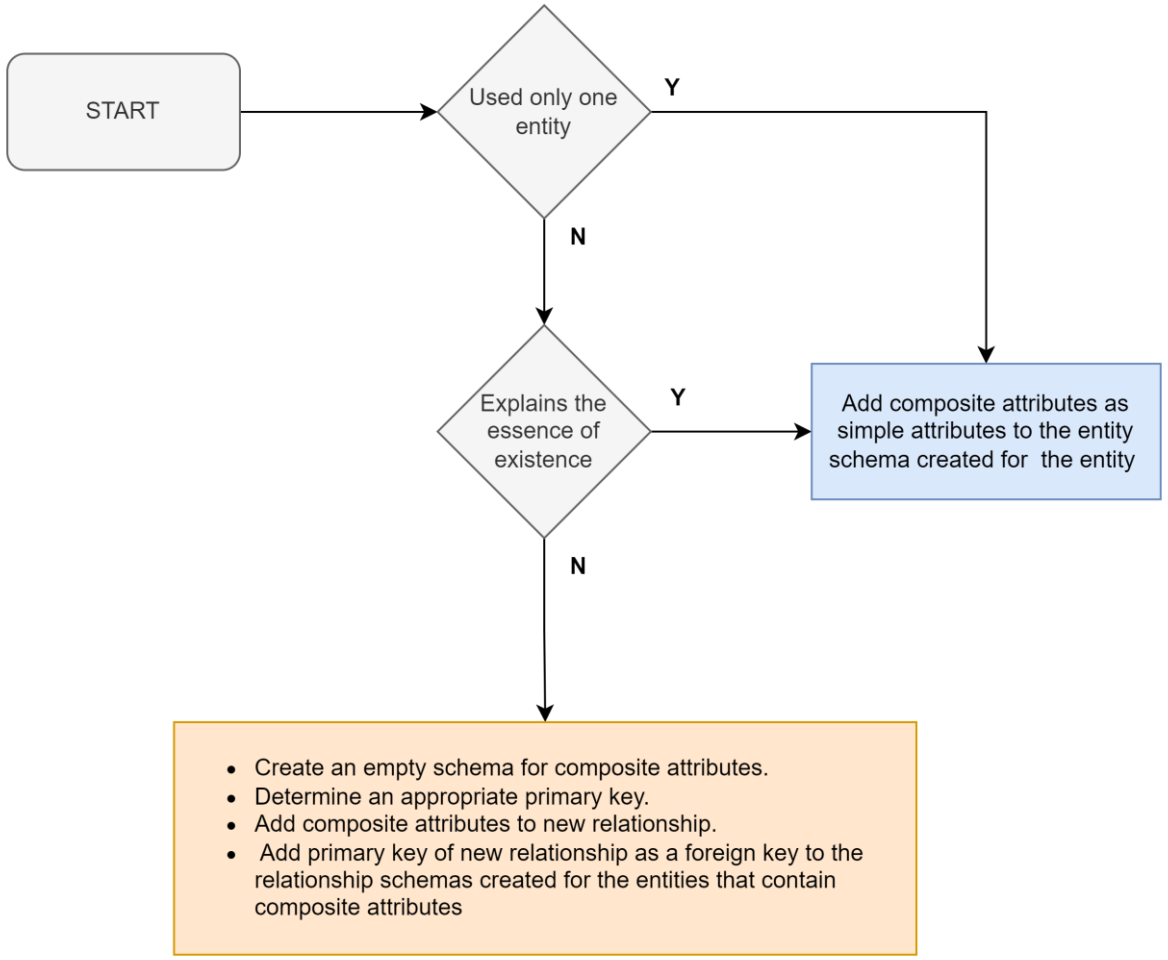
Order

Order_Num	Customer_Code	Ord_Date

Product

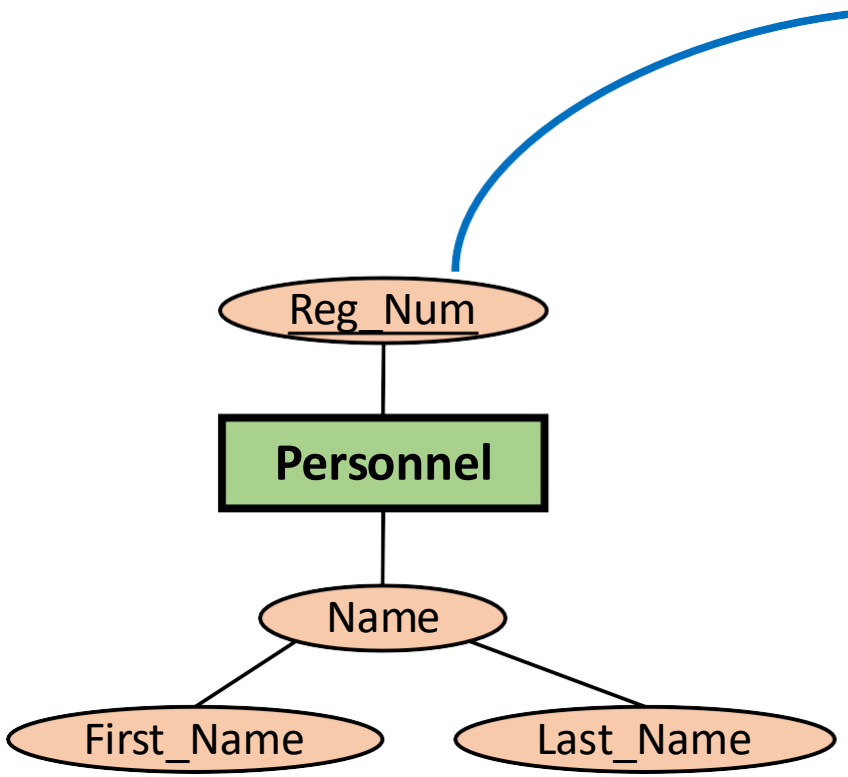
Order_Num	Product_Code	Quantity	Unit_Price	Total_Price=Quantity*Unit_Price

6. Conversion of Composite Attributes



6. Conversion of Composite Attributes

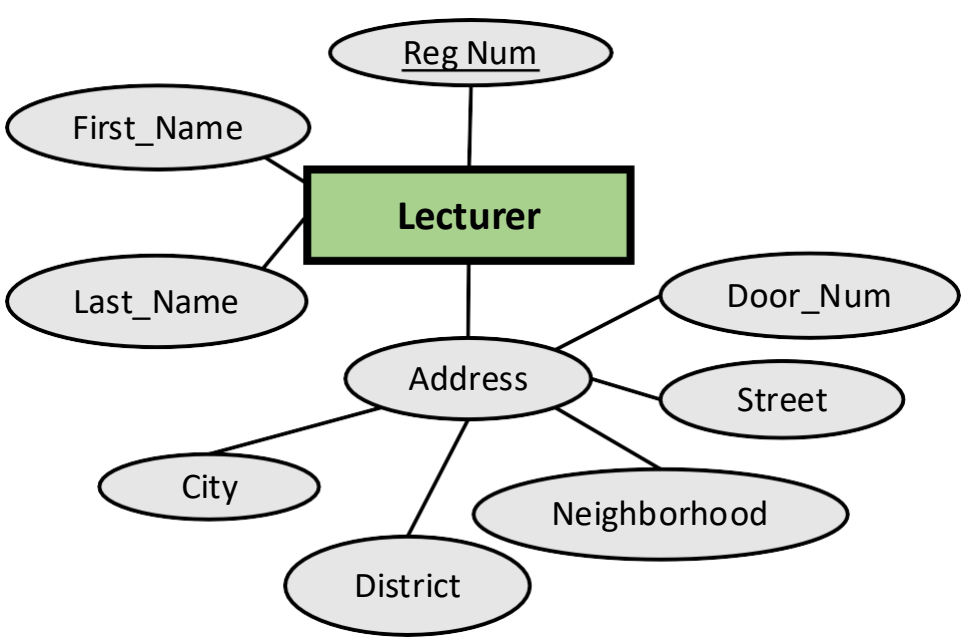
Components describe the entity in first degree.

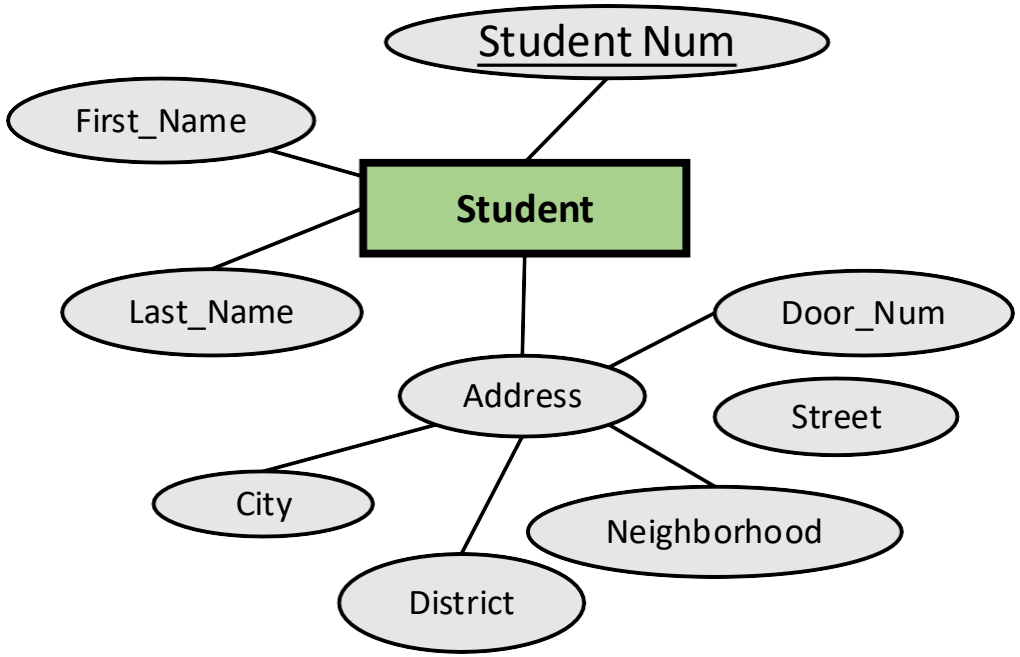


Personnel

Reg_Num	First_Name	Last_Name

6. Conversion of Composite Attributes





6. Conversion of Composite Attributes

Lecturer

Reg_Num	First_Name	Last_Name	Addr_Code

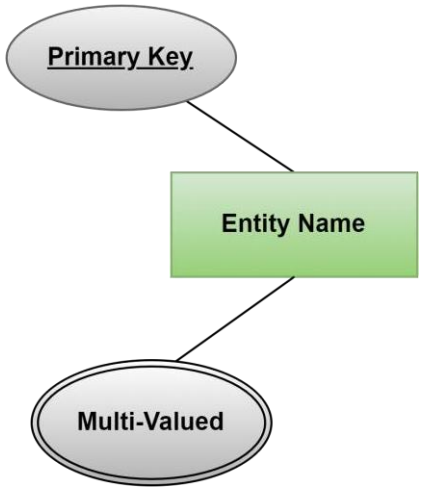
Address

Addr_Code	City	District	Neighborhood	Street	Door_Num

Student

Student_Num	First_Name	Last_Name	Addr_Code

7. Conversion of Multi-Valued Attributes



- Create the relationship schema of the entity using known methods.
- Create an empty schema for multi-valued attribute.
- Add primary key of the entity as a foreign key to the new relation schema. This will be the key part of the new relationship.
- Draw arrow starting from the foreign key and ending at the primary key.
- Add the multi-valued attribute as a key part to the new relationship.

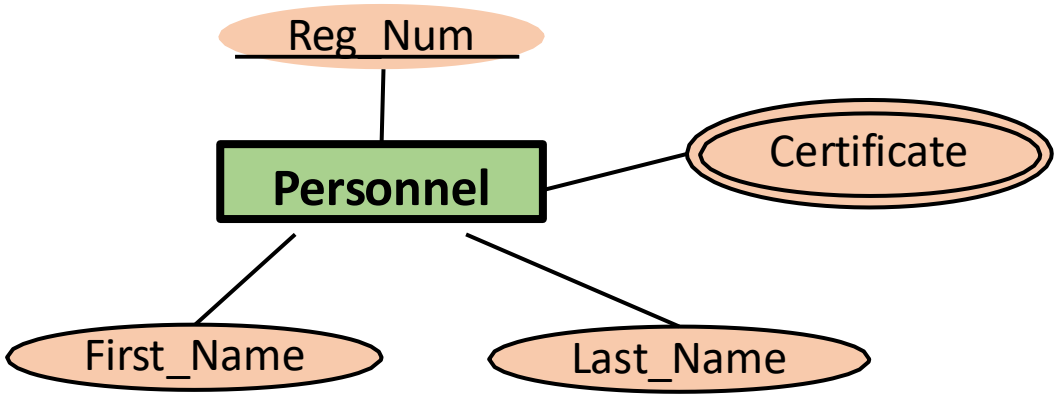
Entity

Primary Key_A	....

New Entity for MV Attr.

Foreign Key_A	MV Attribute

7. Conversion of Multi-Valued Attributes



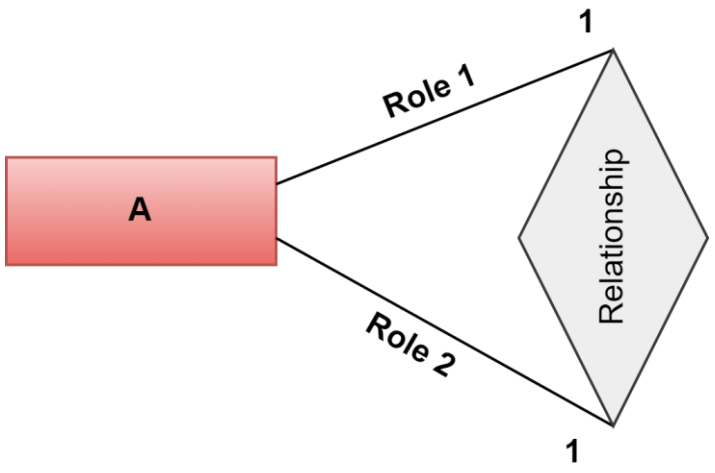
Personnel

Reg_Num	First_Name	Last_Name

Personnel Certificate

Reg_Num	Certificate

8. Conversion of One-to-One Unary Relationships



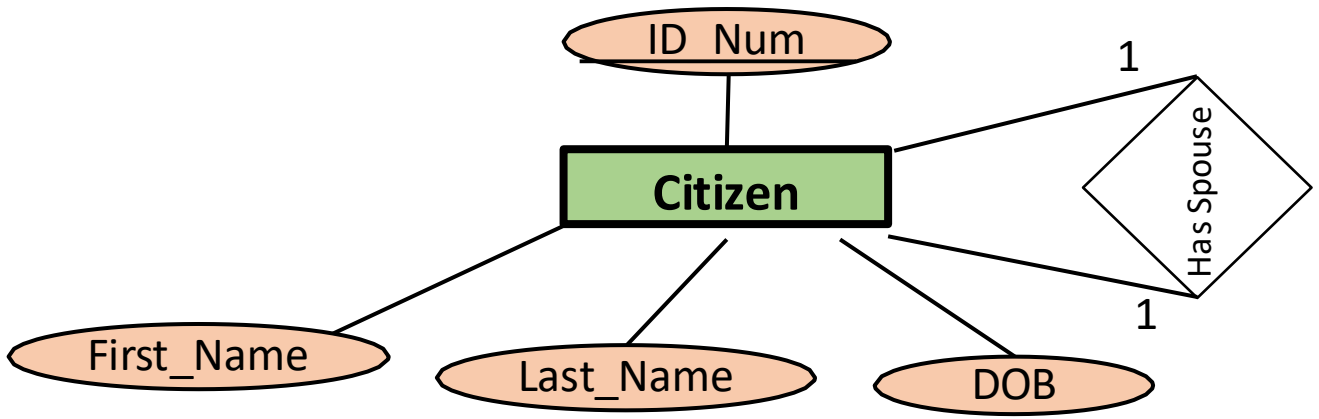
- Create the relationship schema of the entity using known methods.
- Add new attribute has a foreign key is headed to the primary key.
- The attribute name should explain the relationship.

Entity

Primary Key_A	....	Foreign_Key_A

Recursive Foreign Key

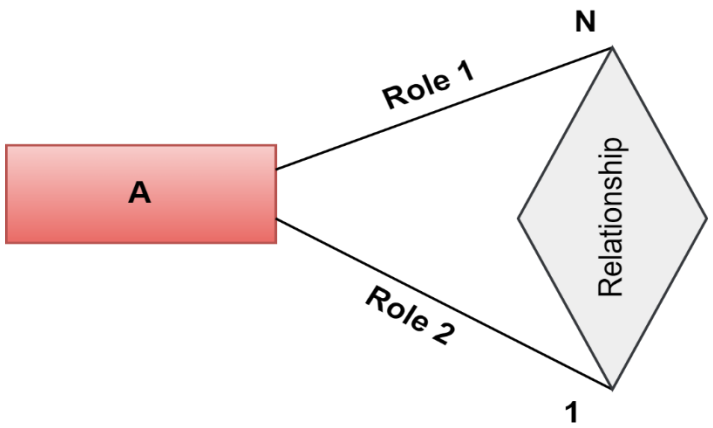
8. Conversion of One-to-One Unary



Citizen

ID_Num	First_Name	Last_Name	DOB	ID_Num_of_Spouse

9. Conversion of One-to-N Unary Relationshhips



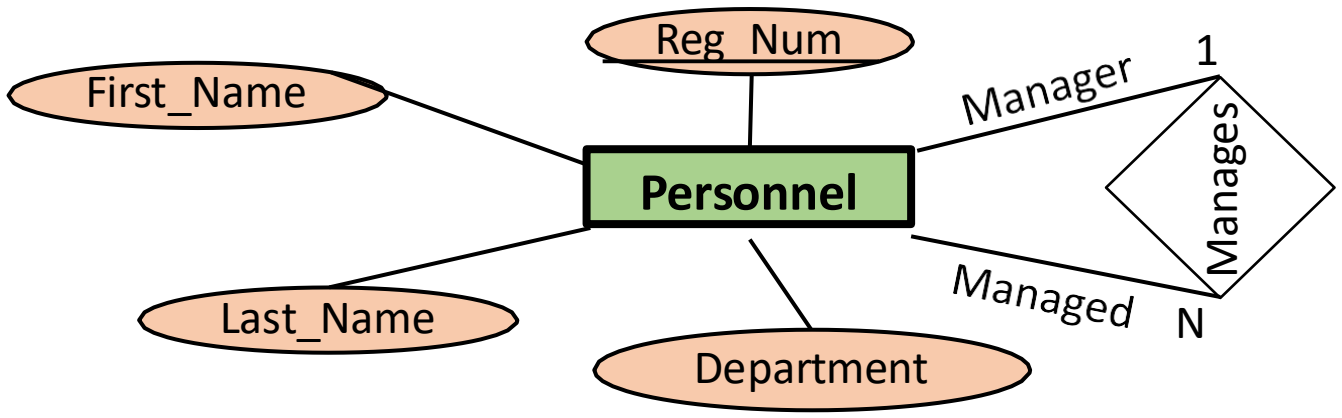
- Can be converted as the same as 1:1 unary relationship.
- The value of the primary key attribute on the "1" side of the relation is kept in the foreign key.
- The difference from the 1:1 relation is executed in operation.

Entity

Primary Key_A	....	Foreign_Key_A

Recursive Foreign Key

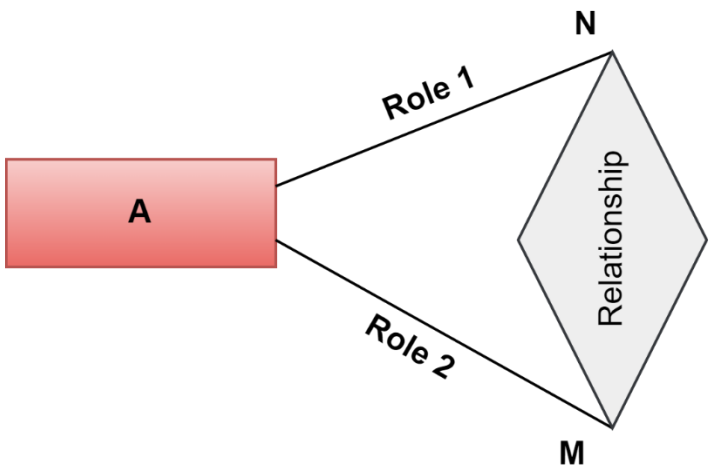
9. Conversion of One-to-N Unary Relationships



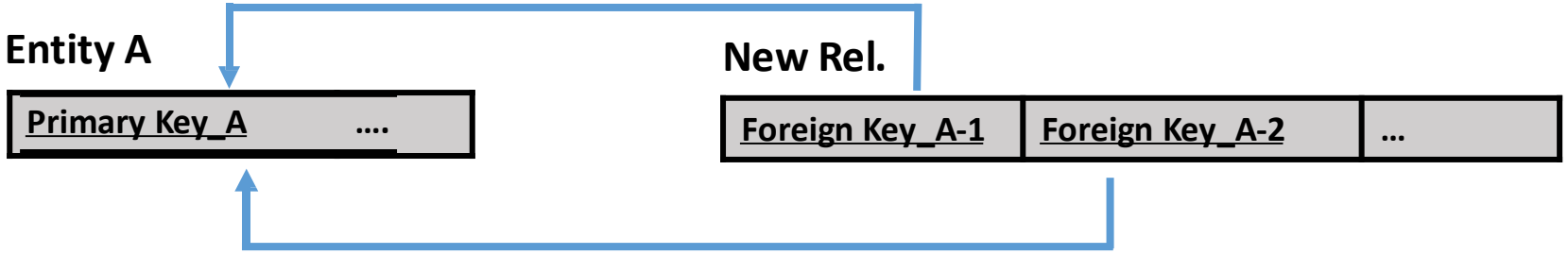
Personnel

Reg_Num	First_Name	Last_Name	Department	Reg_Num_of_Manager

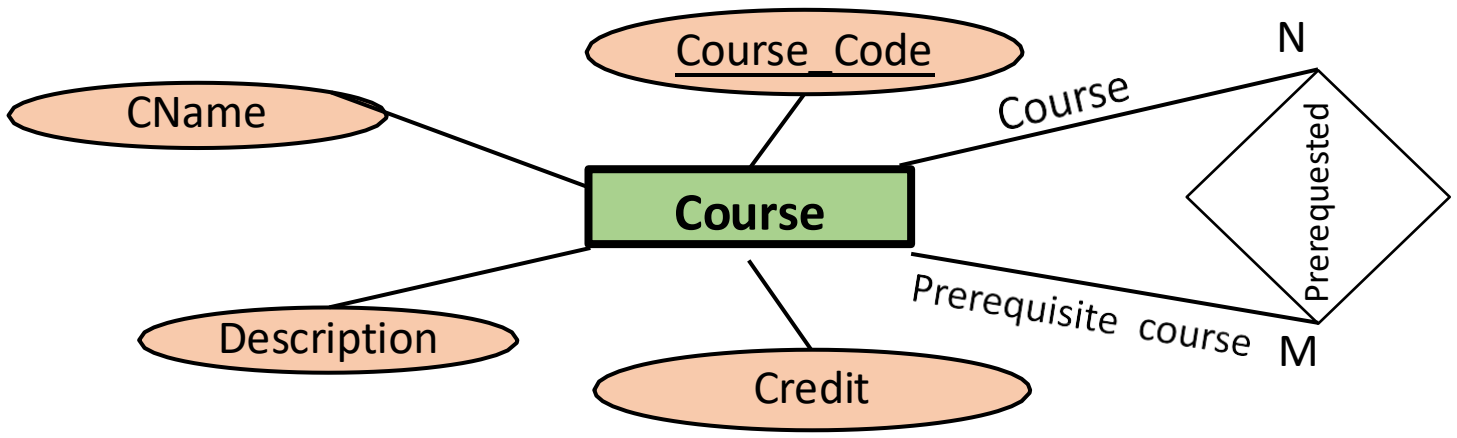
10. Conversion of N-to-M Unary Relationships

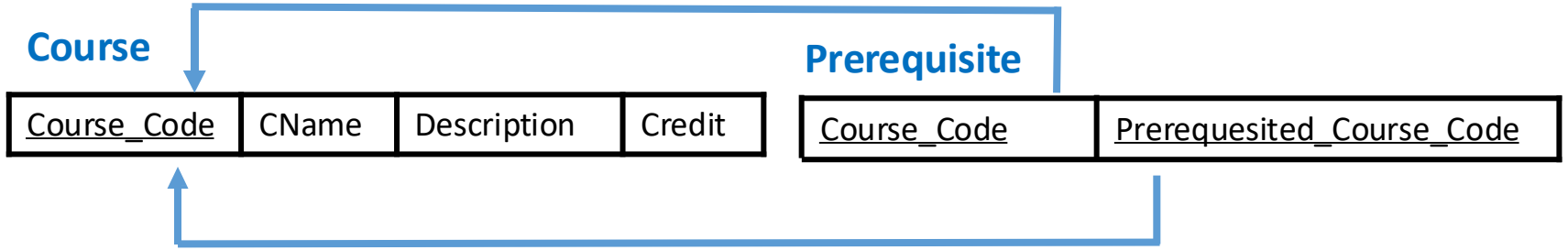


- Create the relationship schema of the entity usign known methods.
- Create an empty schema for the relationship.
- Add primary key of the entity to the new relationship schema twice as foreign keys. These attributes will be the primary key of the new relationship.

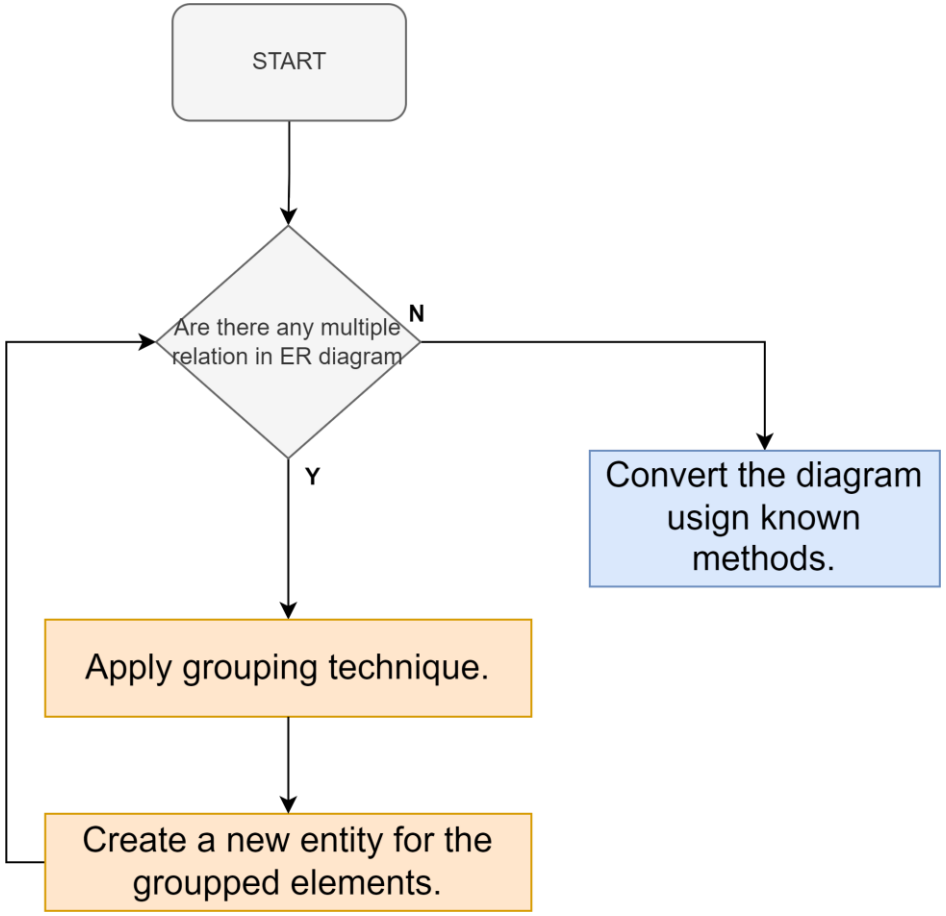


10. Conversion of N-to-M Unary Relationships





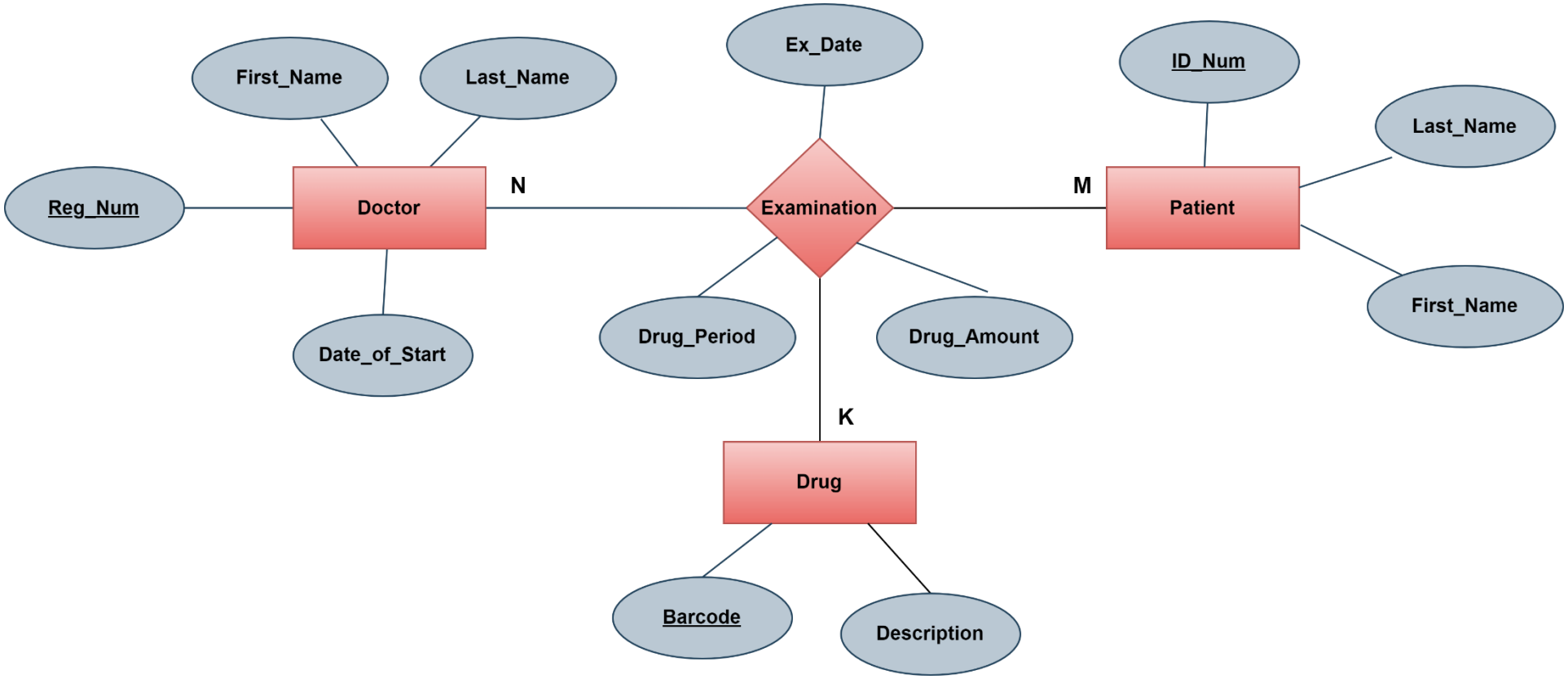
11. Converting Ternary and N-Relations



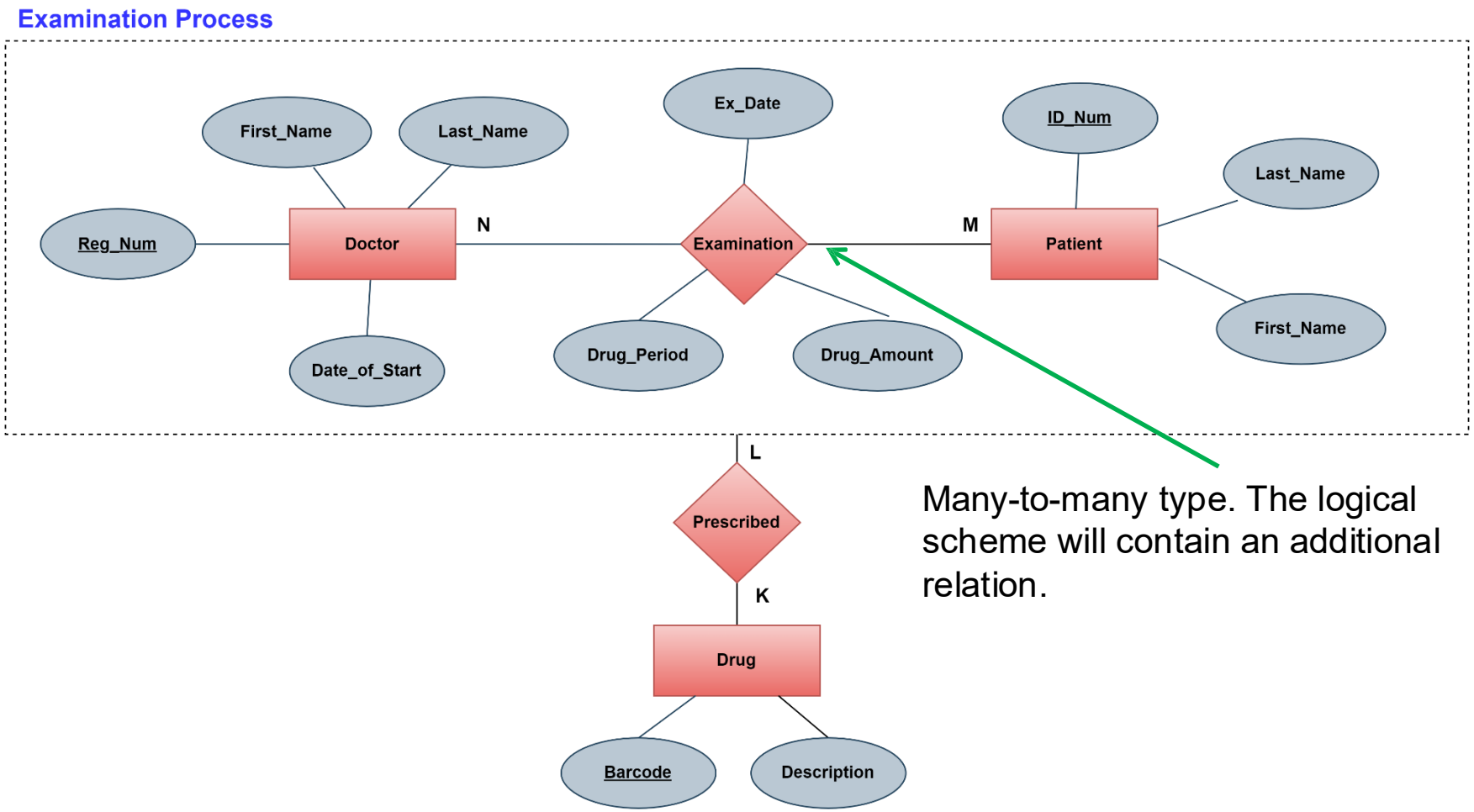
11. Converting Ternary and N-Relations

- Within the scope of the patient examination system:
- A doctor may examine any patient more than once at different times.
- A patient may be examined more than once by the same or different doctor at different times.
- More than one different drug may be prescribed in a specific examination. (for patient)
- A specific drug may be prescribed in more than one examination. (for doctor)
- Drug use information (amount and period) and date of examination should also be followed.
- Information is expected below:
- For Doctor: Registration Number, job start date, First and Last Name.
- For Patient: Citizen ID Number, First and Last Name.
- For Drug: Barcode and Description.

11. Converting Ternary and N-Relations

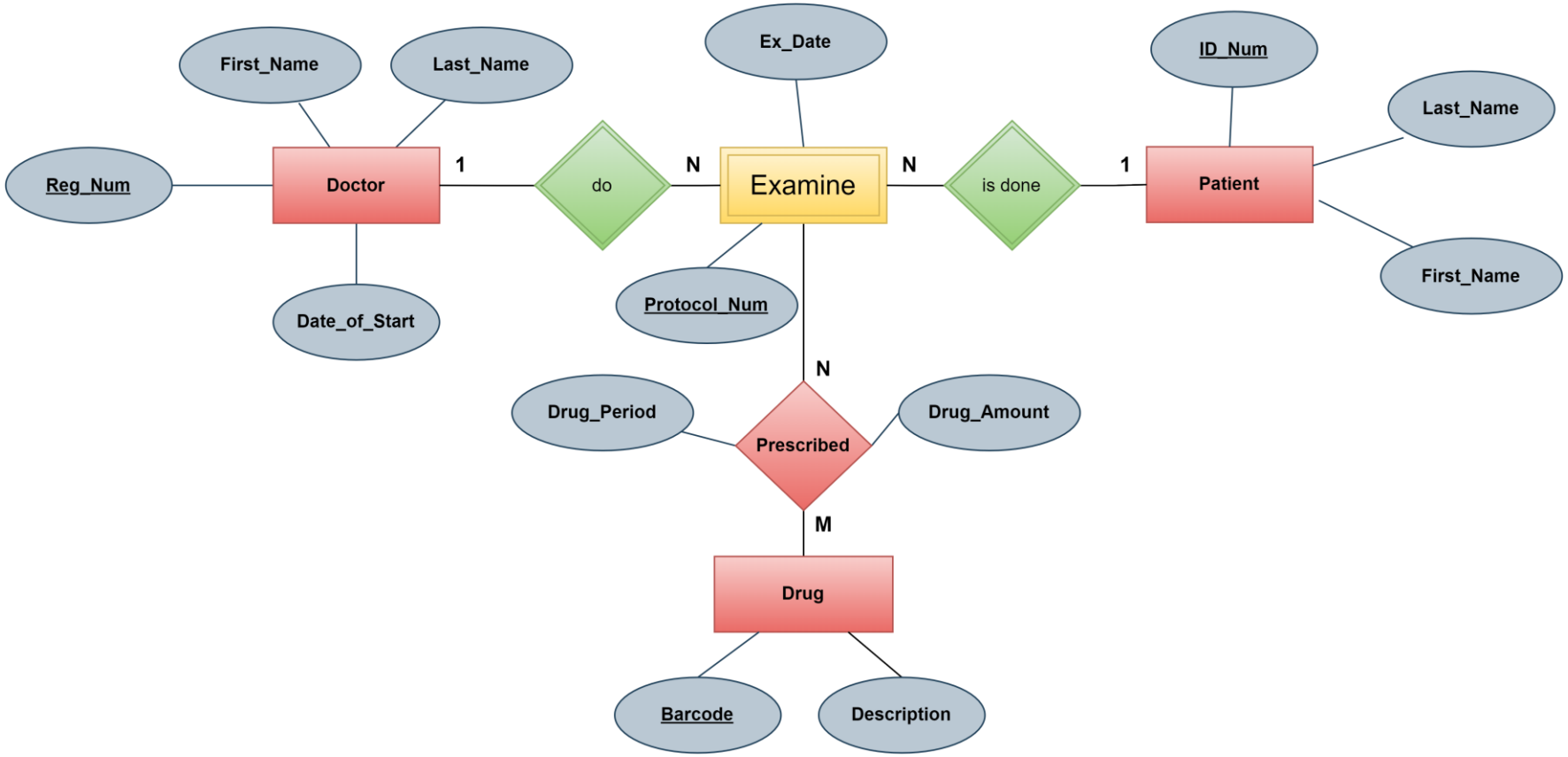


11. Converting Ternary and N-Relations

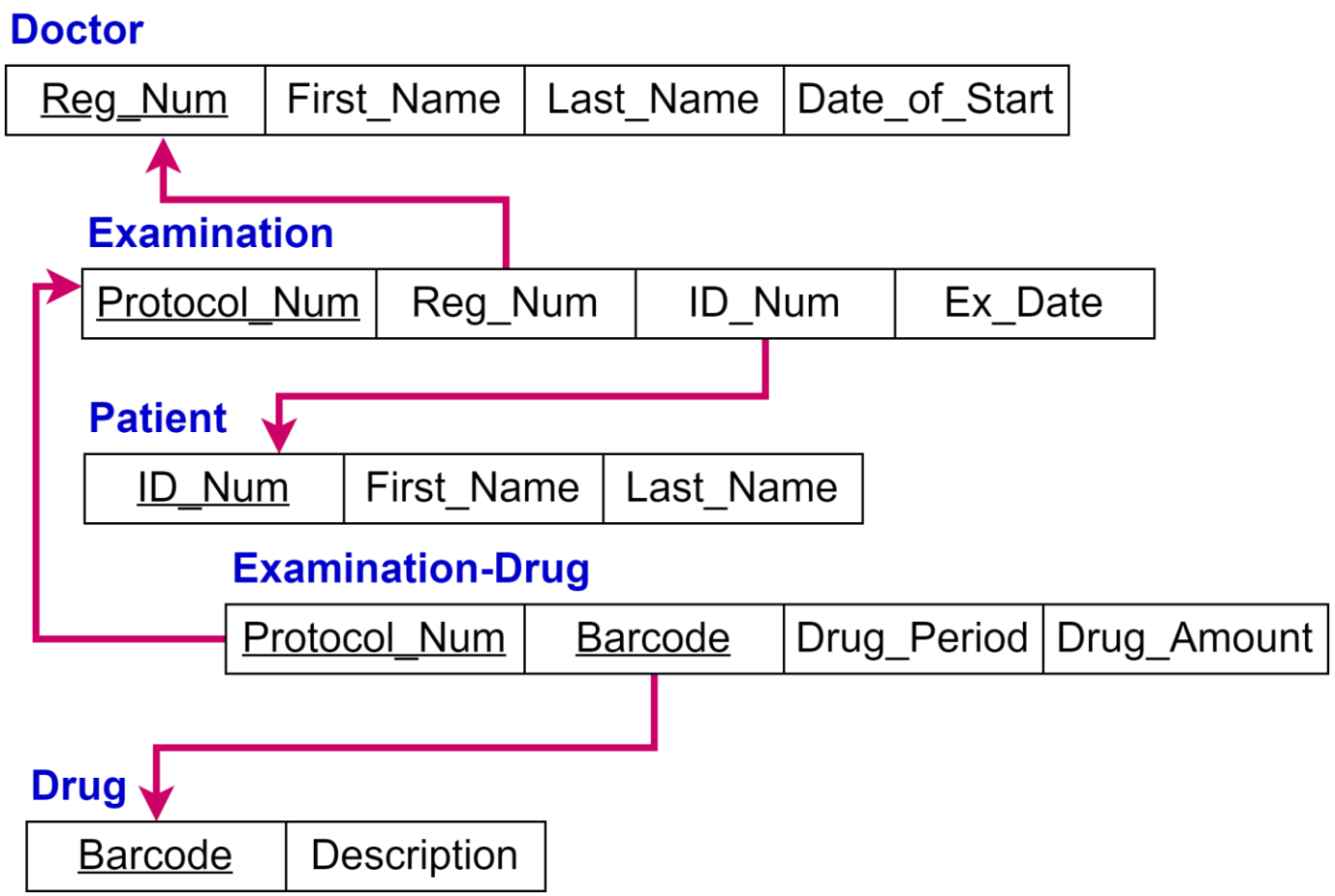


Many-to-many type. The logical scheme will contain an additional relation.

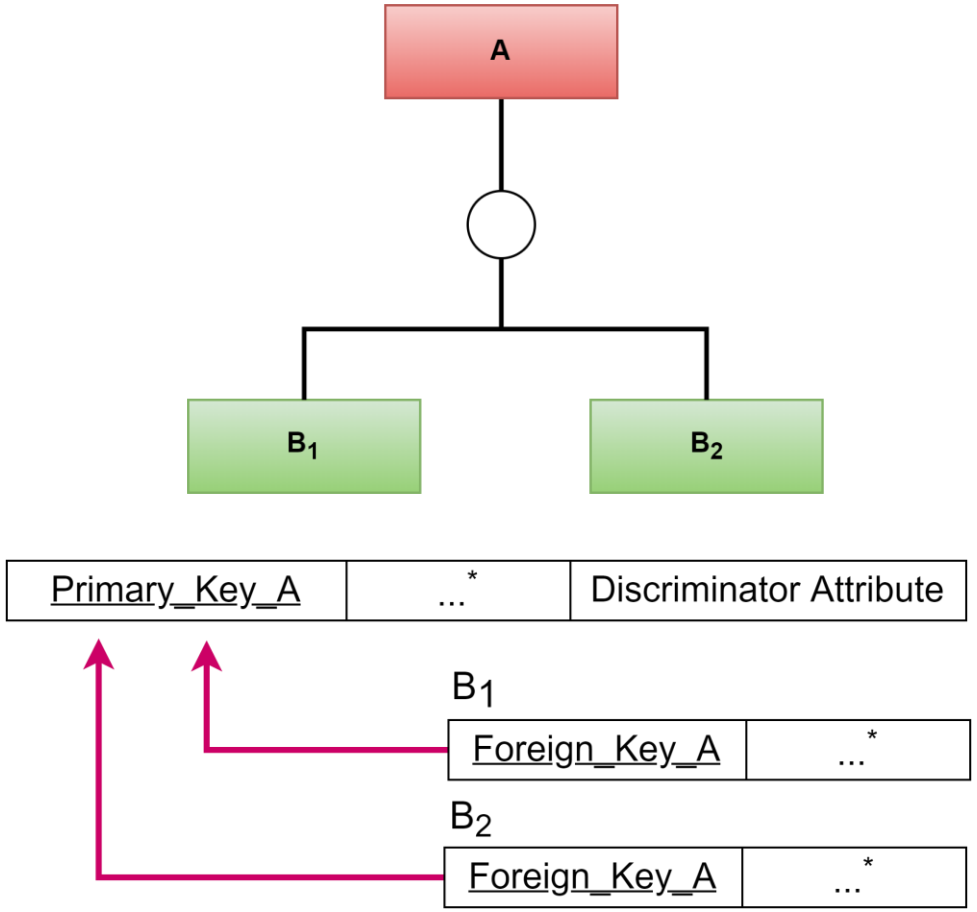
11. Converting Ternary and N-Relations



11. Converting Ternary and N-Relations

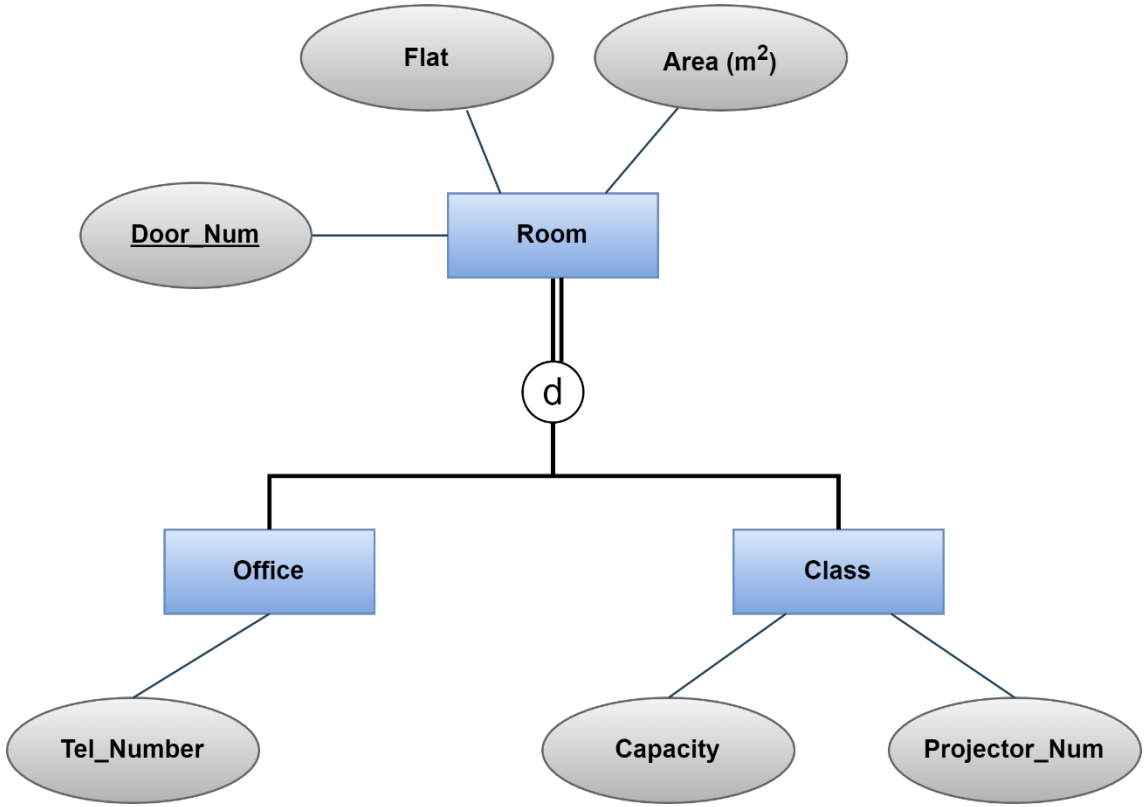


12. Conversion of Sub/Super Class Entities

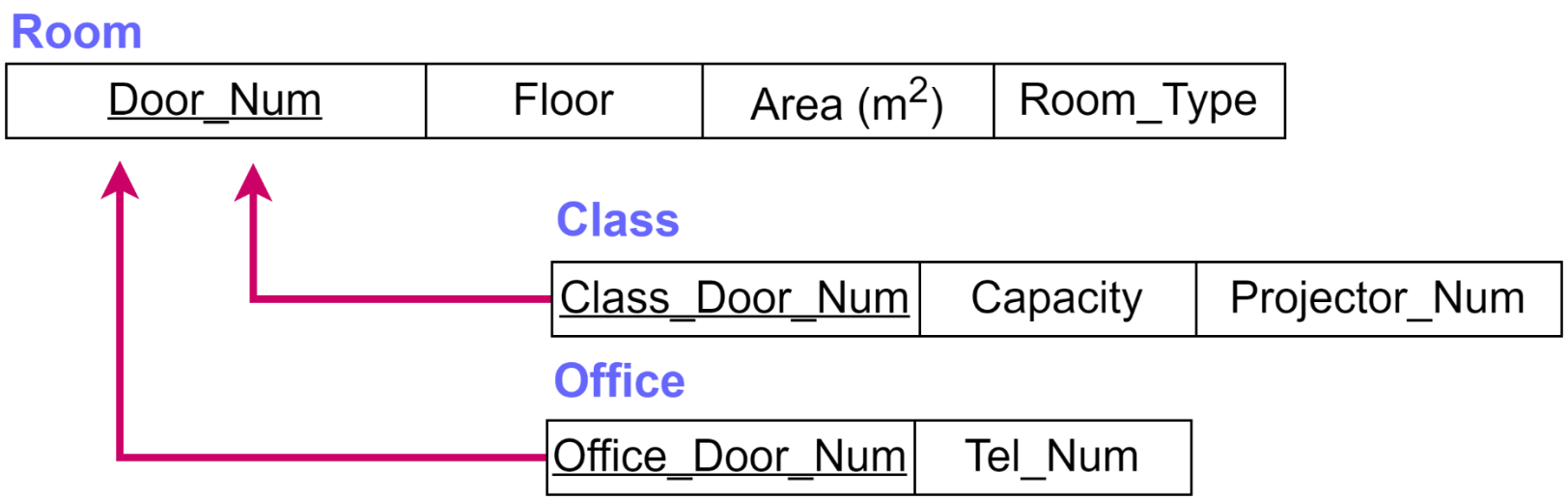


- Create a relationship for the superclass entity that contains its own attributes.
- Create relationships for each subclass entity.
  - Add the primary key of the superclass relationship as a foreign key to these relationships. These keys will also serve as the primary key of the subclass relations.
- Add their own (specific) attributes to these subclass relations.
- Add a discriminative attribute to the relationship created for the superclass.
  - This attribute is used to store the subtype of the entity it contains.

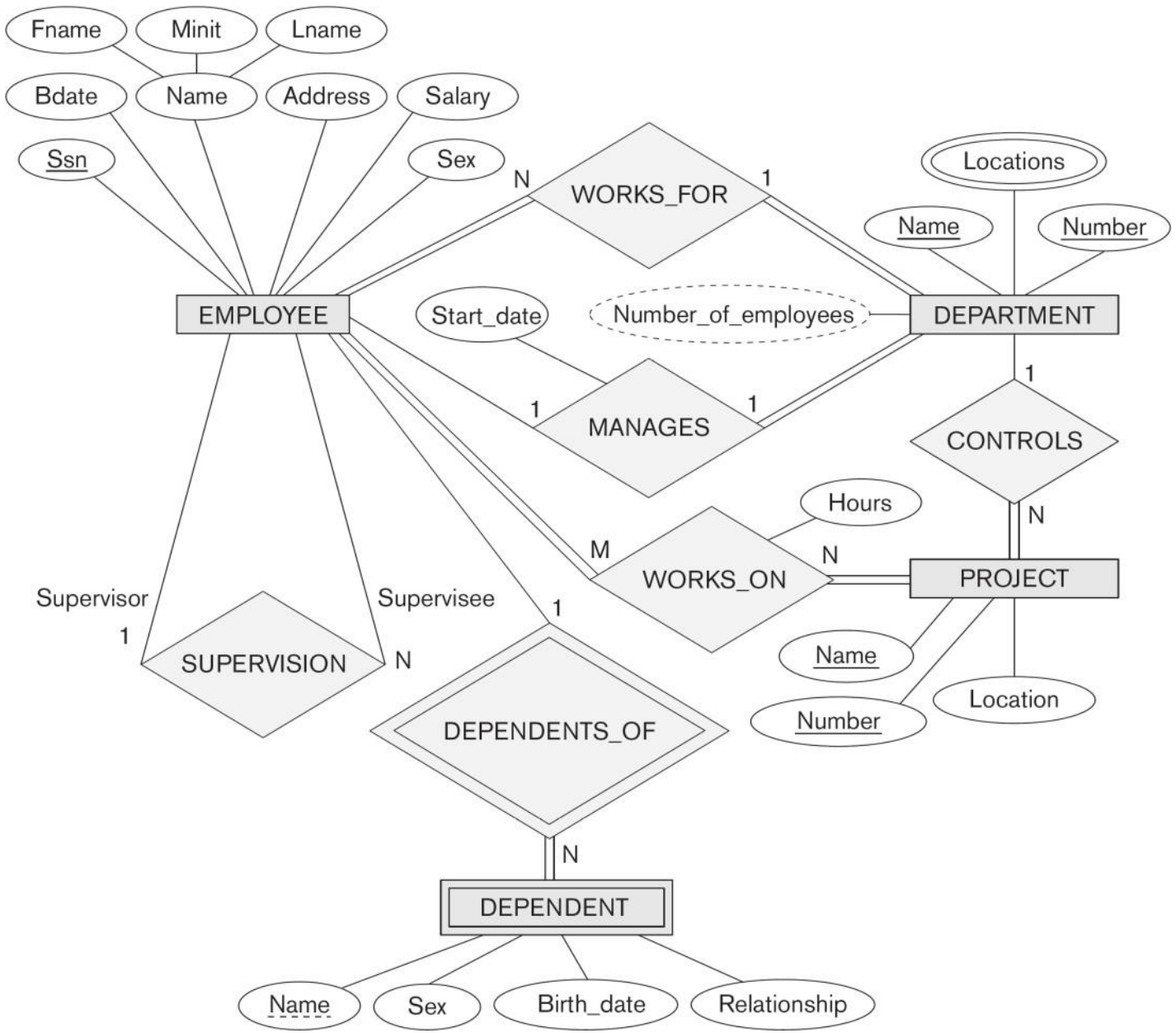
12. Conversion of Sub/Super Class Entities



12. Conversion of Sub/Super Class Entities



Exercise


Fname
Minit
Lname
Bdate
Name
Address
Salary
Ssn
Sex
N
WORKS_FOR
1
Locations
Name
Number
EMPLOYEE
Start_date
Number_of_employees
DEPARTMENT
1
1
MANAGES
1
CONTROLS
Hours
N
M
WORKS_ON
N
PROJECT
Supervisor
Supervisee
1
1
SUPERVISION
N
Name
Location
Number
DEPENDENTS_OF
N
DEPENDENT
Name
Sex
Birth_date
Relationship

Exercise

Step 1: Regular Entity Types

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary

DEPARTMENT

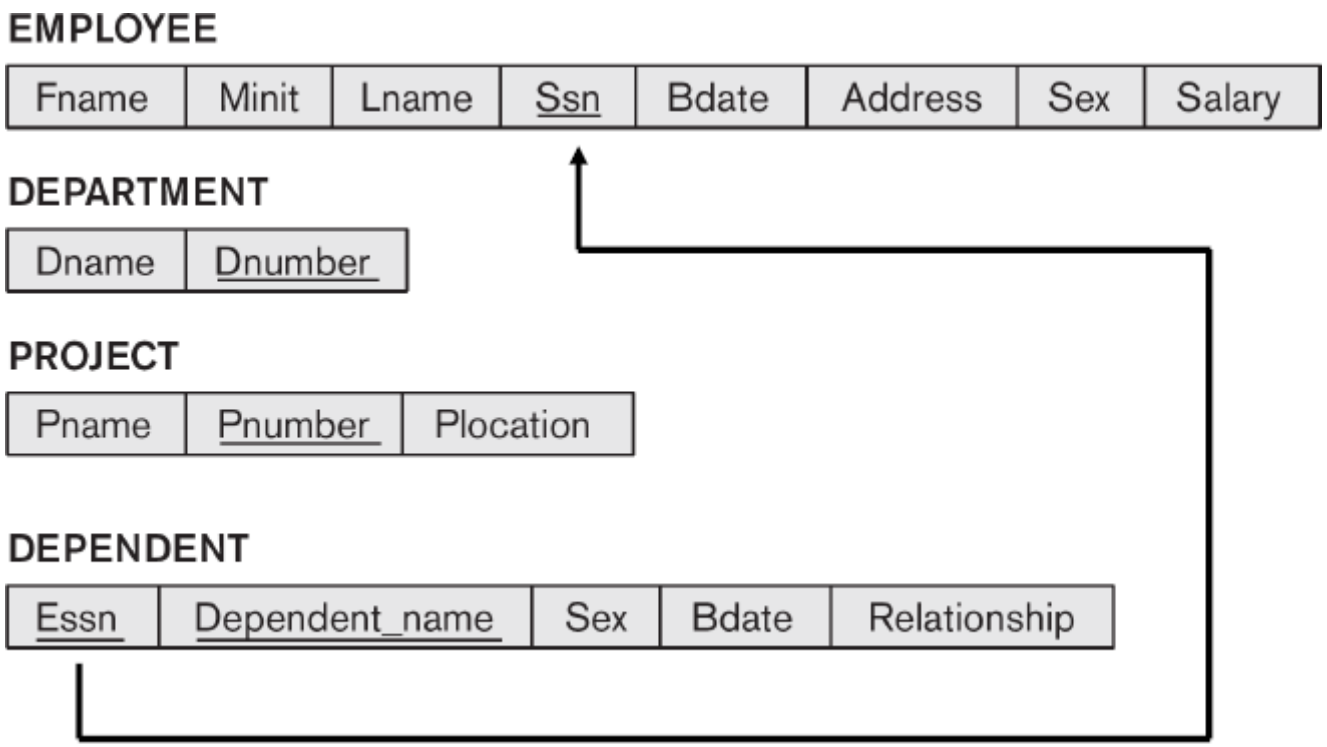
Dname	Dnumber

PROJECT

Pname	Pnumber	Plocation

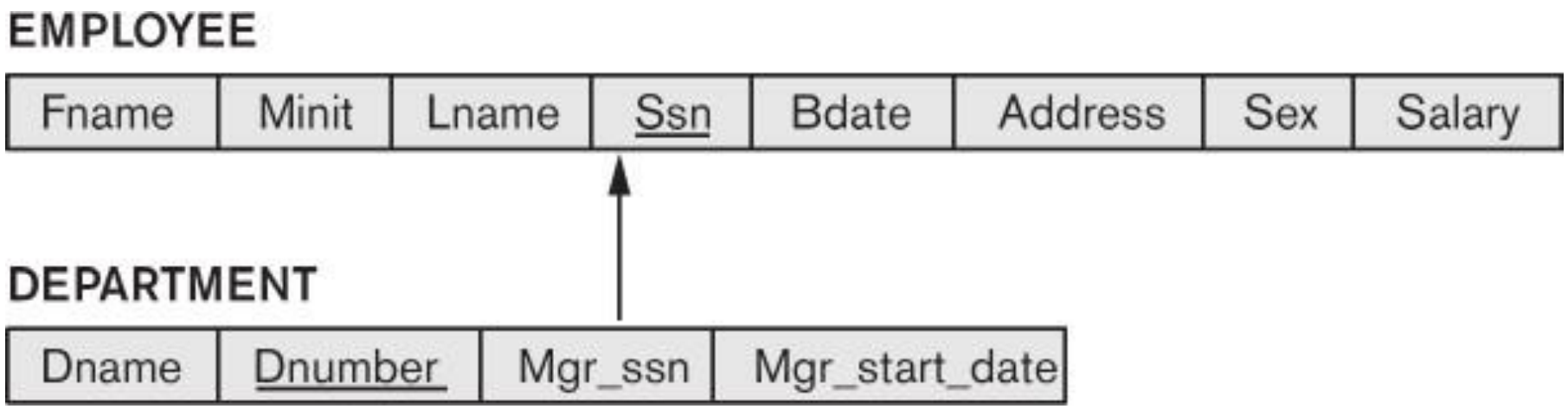
Exercise

Step 2: Weak Entity Types



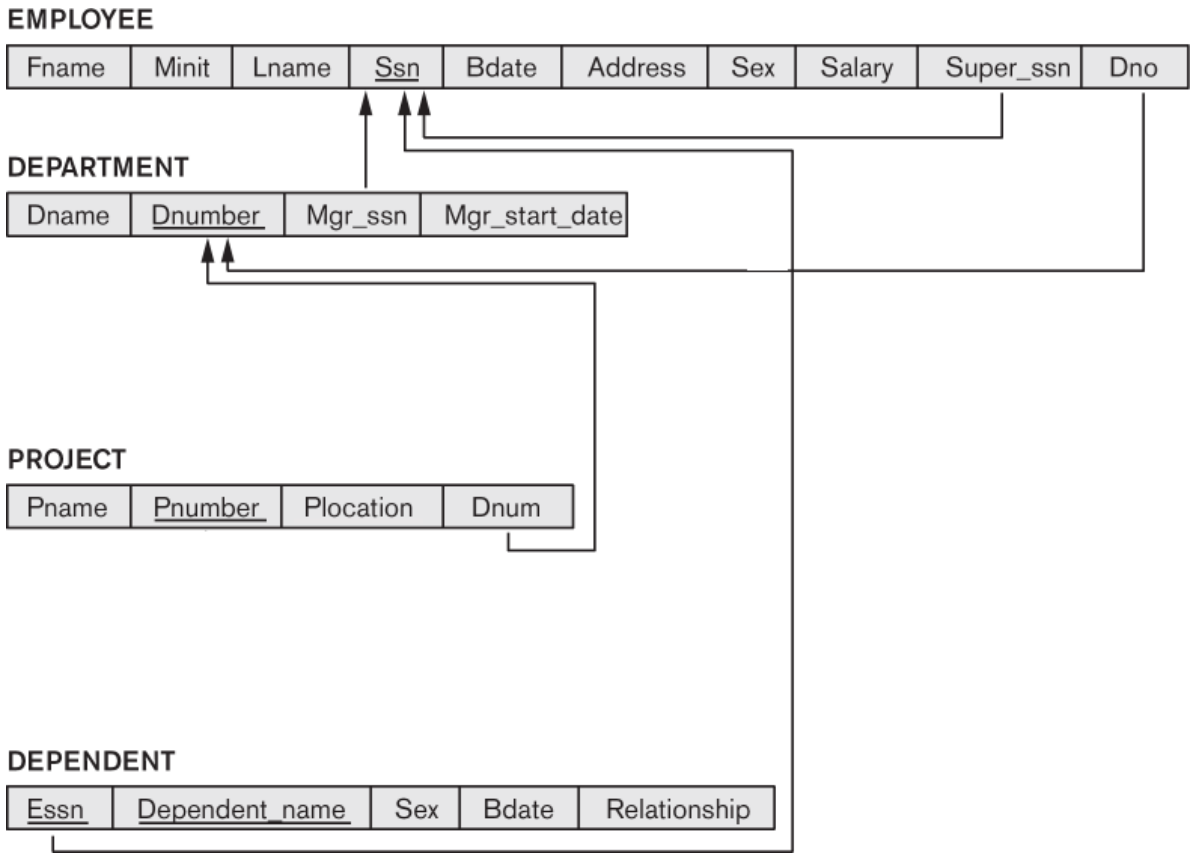
Exercise

Step 3: Mapping Binary 1-to-1



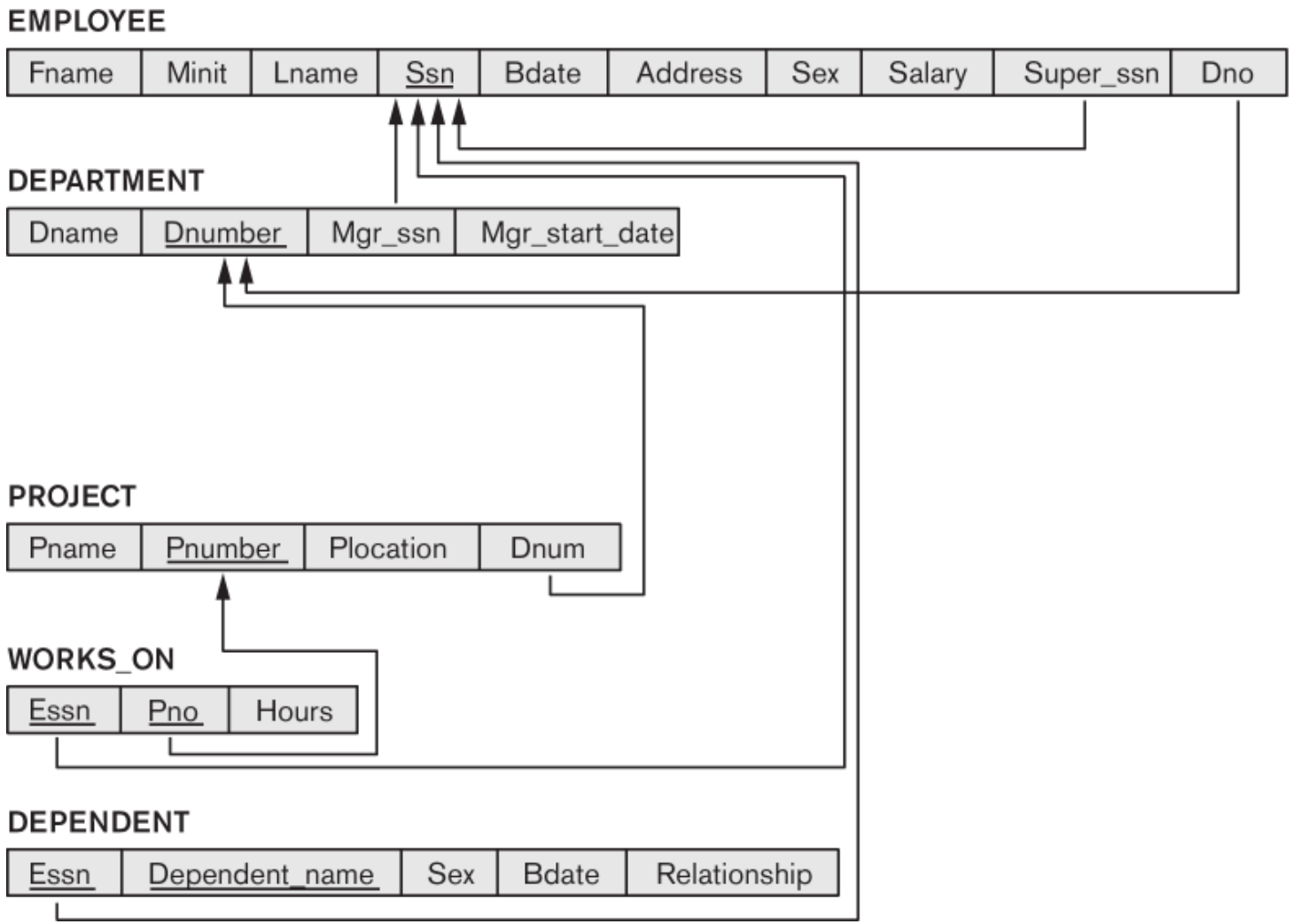
Exercise

Step 4: Binary 1-to-N



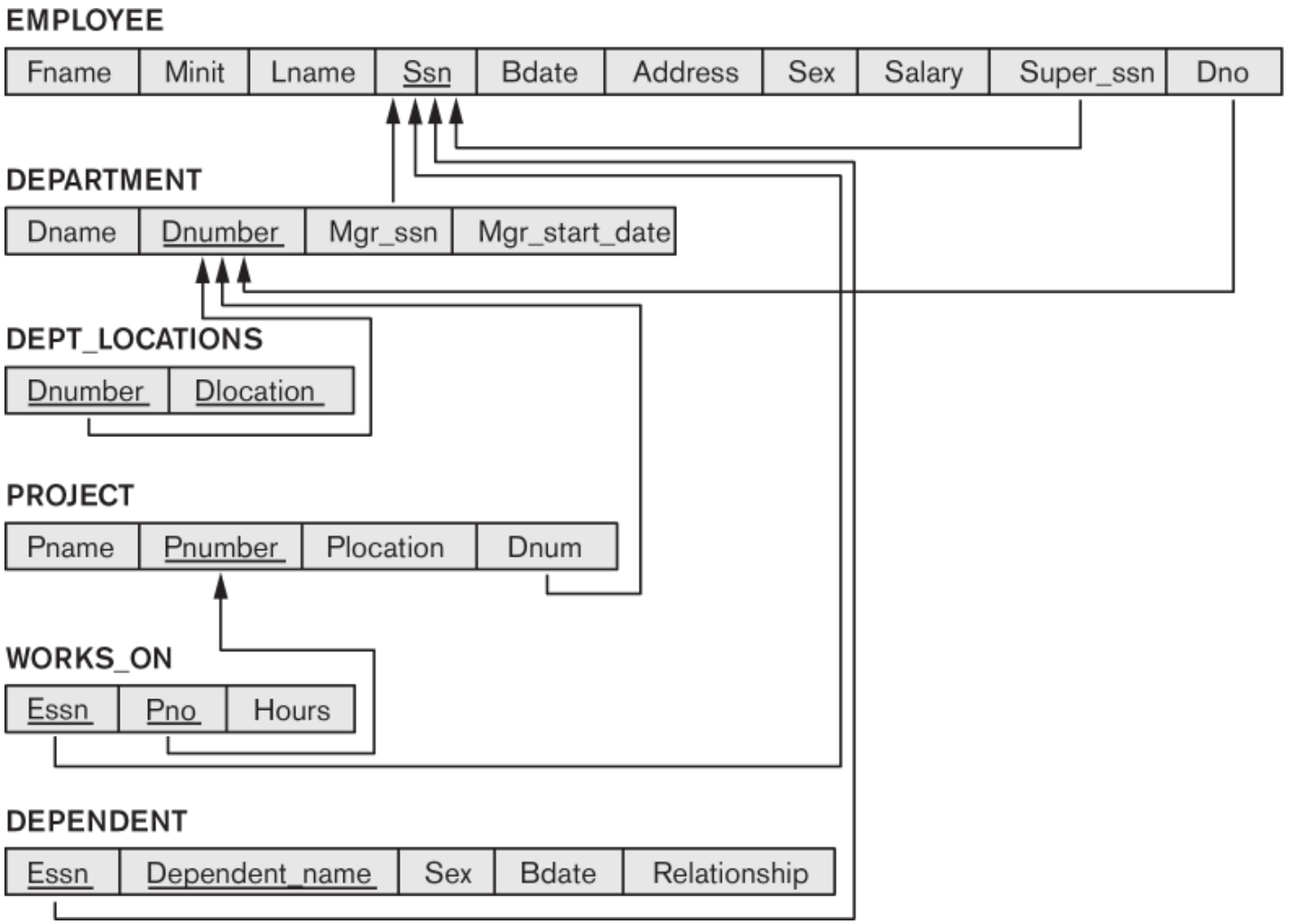
Exercise

Step 5: Binary M-to-N

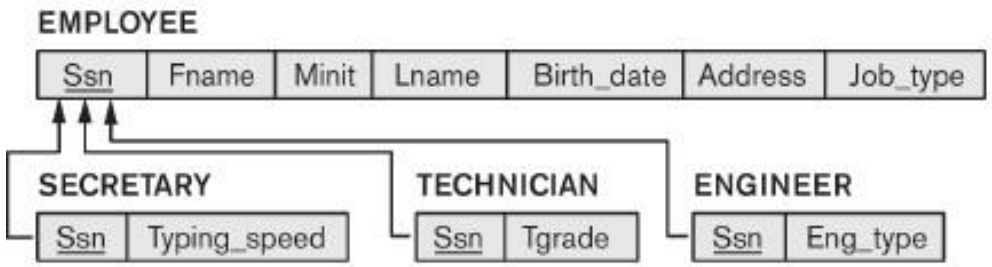


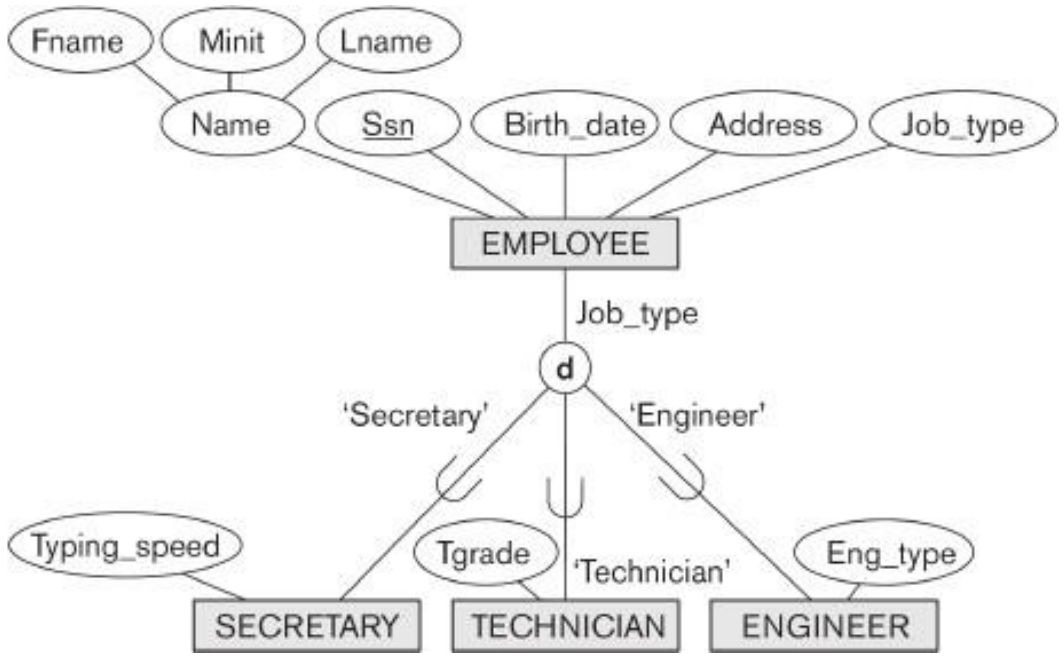
Exercise

Step 6: Multivalued Attributes



Exercise - Specialization/Generalization





THE END

CHECK OUT COURSE WEBSITE: HTTPS://GAZI-END-MUH.GITHUB.IO

GOT ANY QUESTIONS LATER?

SEND ME AN EMAIL: DR.ERCAN.EZIN@GMAIL.COM